QUESTION ANALYSIS

# EEE421

Solid State Devices

# Question Analysis

**Sessions 2021–22 to 2018–19** *4 papers, 109 questions, 148 formulas*

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# 1. At a glance:

This analysis covers **4 papers** (2021-22, 2020-21, 2019-20, 2018-19), with 109 questions and 566 marks offered. It shows what past papers asked. It does not predict the next paper, and with so few papers a pattern can easily be chance.

### The five topics with the most marks:

| Topic | Marks | Share | Questions | Top subtopics (marks) |
|---|---|---|---|---|
| Semiconductor in equilibrium | 130 | 23 % | 32 | Carrier concentrations (56); Compensation (19); Ionization (17) |
| Excess carriers | 78 | 14 % | 13 | Steady state (29); Ambipolar transport (14); Dielectric relaxation (7) |
| Carrier transport | 75 | 13 % | 12 | Hall effect (25); Mobility (25); Einstein relation (11) |
| PN junction | 72 | 13 % | 12 | Depletion width (25); Ideal diode current (15); Biased junction (10) |
| Quantum basics | 58 | 10 % | 13 | Fermi probability (45); Density of states (7); Effective mass (3) |

In the topic pages, a formula name in bold is a key formula, the number after a subtopic is its marks offered, and a blue symbol in an equation links to its meaning in the symbol list, where the arrow leads back to the equation.

# 2. Heat map:

Marks offered, by topic and paper, in book order. A darker cell means more marks. Click a topic to open its page.

| Topic and subtopic | 2021-22 | 2020-21 | 2019-20 | 2018-19 | Term tests | All |
|---|---|---|---|---|---|---|
| **Crystal structure** | · | · | **12** | · | · | **12** |
| Classes of solids | · | · | **6** | · | · | 6 |
| Unit cell | · | · | **6** | · | · | 6 |
| **Quantum basics** | **8** | **15** | · | **19** | **16** | **58** |
| Fermi probability | **8** | **10** | · | **11** | **16** | 45 |
| Density of states | · | **5** | · | **2** | · | 7 |
| Effective mass | · | · | · | **3** | · | 3 |
| Energy bands | · | · | · | **3** | · | 3 |
| **Semiconductor in equilibrium** | **12** | **30** | **25** | **16** | **47** | **130** |
| Carrier concentrations | **5** | **5** | **6** | **9** | **31** | 56 |
| Compensation | **7** | **5** | **6** | · | **1** | 19 |
| Ionization | · | **15** | · | · | **2** | 17 |
| Fermi level | · | **5** | · | · | **10** | 15 |
| Temperature dependence | · | · | **5** | **7** | · | 12 |
| Carrier distribution | · | · | **8** | · | · | 8 |
| Dopants | · | · | · | · | **2** | 2 |
| Degenerate | · | · | · | · | **1** | 1 |
| **Carrier transport** | · | **25** | **27** | **20** | **3** | **75** |
| Hall effect | · | **10** | **8** | **7** | · | 25 |
| Mobility | · | **10** | **12** | · | **3** | 25 |
| Einstein relation | · | **5** | · | **6** | · | 11 |
| Diffusion | · | · | · | **7** | · | 7 |
| Thermal velocity | · | · | **7** | · | · | 7 |
| **Excess carriers** | · | **20** | **31** | **6** | **21** | **78** |

| Topic and subtopic | 2021-22 | 2020-21 | 2019-20 | 2018-19 | Term tests | All |
|---|---|---|---|---|---|---|
| Steady state | · | **10** | **7** | **6** | **6** | 29 |
| Ambipolar transport | · | · | **7** | · | **7** | 14 |
| Dielectric relaxation | · | · | **7** | · | · | 7 |
| Generation and recombination | · | · | · | · | **7** | 7 |
| Time and space dependence | · | · | **6** | · | · | 6 |
| Minority carriers | · | **5** | · | · | · | 5 |
| Carrier pulse | · | **5** | · | · | · | 5 |
| Quasi-Fermi levels | · | · | **4** | · | · | 4 |
| Excess carriers | · | · | · | · | **1** | 1 |
| **PN junction** | · | **25** | **24** | **20** | **3** | **72** |
| Depletion width | · | **10** | **12** | · | **3** | 25 |
| Ideal diode current | · | · | **8** | **7** | · | 15 |
| Biased junction | · | **10** | · | · | · | 10 |
| Switching transient | · | · | · | **7** | · | 7 |
| Junction capacitance | · | · | · | **6** | · | 6 |
| Band diagram | · | **5** | · | · | · | 5 |
| Built-in potential | · | · | **4** | · | · | 4 |
| **Tunnel diode** | · | **5** | · | **6** | · | **11** |
| Operation | · | **5** | · | **6** | · | 11 |
| **Metal-semiconductor contacts** | · | · | **6** | **10** | · | **16** |
| Schottky and ohmic contacts | · | · | **6** | **10** | · | 16 |
| **MOS structure** | · | **15** | **15** | **24** | · | **54** |
| Threshold voltage | · | · | **12** | **6** | · | 18 |
| Band diagrams | · | **5** | · | **8** | · | 13 |
| Body effect | · | · | · | **5** | · | 5 |
| C-V characteristics | · | · | · | **5** | · | 5 |
| Double gate | · | **5** | · | · | · | 5 |
| Inversion | · | **5** | · | · | · | 5 |
| MOSFET advantages | · | · | **3** | · | · | 3 |
| **Bipolar transistor** | · | **10** | **10** | **27** | · | **47** |
| Operating modes | · | **5** | **6** | · | · | 11 |
| Emitter efficiency | · | · | **4** | **6** | · | 10 |
| Current crowding | · | · | · | **7** | · | 7 |
| Amplification | · | · | · | **6** | · | 6 |
| Base width modulation | · | · | · | **5** | · | 5 |
| Minority distribution | · | **5** | · | · | · | 5 |
| Base current | · | · | · | **3** | · | 3 |
| **JFET** | · | · | · | **13** | · | **13** |
| Pinch-off | · | · | · | **7** | · | 7 |
| Frequency response | · | · | · | **6** | · | 6 |

# 3. Priority list:

**Must know:** Semiconductor in equilibrium (130), Quantum basics (58)

**Likely:** Excess carriers (78), Carrier transport (75), PN junction (72), MOS structure (54), Bipolar transistor (47), Metal-semiconductor contacts (16), Tunnel diode (11)

**Less often:** JFET (13), Crystal structure (12)


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**The rule:** Must know means asked in every paper. Likely means asked in at least half the papers. Less often means asked in fewer than half. Numbers in brackets are marks offered, and each list runs from most marks to fewest.

# 4. Topic pages:

## 4.1. Crystal structure:

12 marks, 2 questions, in 1 of 4 papers. Book: Neamen Ch. 1.

Most often theory (1 of 2). Subtopics: Classes of solids; Unit cell.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Unit cell (6) | **Atoms per fcc cell** | $n = 8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$ | 19-20 1(b) |
| | **Volume density** | $\text{density} = \frac{n}{a^3}$ | 19-20 1(b) |

### Symbols:

$a$ = lattice constant in cm ↩

### Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Classes of solids** | | | | |
| 1 | 19-20 1(a) | 6 | theory | Describe three classifications of solids-amorphous, polycrystalline, and single crystal. |
| **Unit cell** | | | | |
| 2 | 19-20 1(b) | 6 | numerical | The lattice constant of a face-centered cubic lattice is 0.425 nm. Determine the (i) effective number of atoms per unit cell (ii) volume density of atoms |

## 4.2. Quantum basics:

58 marks, 13 questions, in 4 of 4 papers. Book: Neamen Ch. 3.

Most often theory (7 of 13). Subtopics: Fermi probability; Density of states; Effective mass; Energy bands.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Fermi probability (45) | **Fermi-Dirac function** | $f_F(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{kT}\right)}$ | 21-22 7(a), 20-21 8(a), 19-20 7(d), 18-19 1(c), 18-19 2(b) |
| | Thermal energy | $kT = 0.0259$ eV at 300 K | 18-19 2(b) |
| | **Electron concentration** | $n_0 = N_c \exp\left[-\frac{E_c - E_F}{kT}\right]$ | 20-21 8(h), 19-20 7(d) |
| | **Mass-action law** | $n_0 p_0 = n_i^2 = N_c N_v \exp\left(-\frac{E_g}{kT}\right)$ | 19-20 7(d) |

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| | Energy for a given probability | $E - E_F = kT \ln\left(\frac{1}{f_F} - 1\right)$ | 21-22 7(d), 20-21 2(d), 20-21 8(g) |
| | Ways to place electrons | $W_i = \frac{g_i!}{N_i!\,(g_i - N_i)!}$ | 21-22 7(a) |
| | Empty-state probability | $1 - f_F(E) = \frac{1}{1 + \exp\left(\frac{E_F - E}{kT}\right)}$ | 21-22 7(a) |
| | **Boltzmann approximation** | $f_F(E) \approx \exp\left[-\frac{E - E_F}{kT}\right]$ | 21-22 7(a) |
| Density of states (7) | **Density of states** | $g_c(E) = \frac{4\pi(2m_n^*)^{3/2}}{h^3}\sqrt{E - E_c},\ g_v(E) = \frac{4\pi(2m_p^*)^{3/2}}{h^3}\sqrt{E_v - E}$ | 18-19 7(a) |
| | **Electron concentration** | $n_0 = \int_{E_c}^{\infty} g_c(E) f_F(E)\,dE = N_c \exp\left[-\frac{E_c - E_F}{kT}\right]$ | 18-19 7(a) |
| | Hole concentration | $p_0 = N_v \exp\left[-\frac{E_F - E_v}{kT}\right]$ | 18-19 7(a) |
| | **Density of states** | $g_c(E) = \frac{4\pi(2m_n^*)^{3/2}}{h^3}\sqrt{E - E_c}$ | 20-21 1(c) |
| | States from $E_c$ to $E_c + kT$ | $N = \frac{4\pi(2m_n^*)^{3/2}}{h^3} \cdot \frac{2}{3}(kT)^{3/2}$ | 20-21 1(c) |
| Effective mass (3) | **Effective mass** | $\frac{1}{m_n^*} = \frac{1}{\hbar^2}\frac{d^2E}{dk^2}$ | 18-19 7(c) |
| Energy bands (3) | **Bloch function** | $\psi(x) = u(x)e^{jkx}$ | 18-19 7(d) |
| | **Kronig-Penney condition** | $P'\frac{\sin\alpha a}{\alpha a} + \cos\alpha a = \cos ka,\ \alpha^2 = \frac{2mE}{\hbar^2}$ | 18-19 7(d) |


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| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|

## Symbols:

$f_F$ = probability that a state at energy $E$ holds an electron ↩

$E_F$ = Fermi level ↩

$N_c$ = effective density of states in the conduction band ↩

$g_i$ = states ↩

$N_i$ = electrons in level $i$ ↩

$g_c$ = states per unit energy per unit volume ↩

## Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Fermi probability** | | | | |
| 1 | 21-22 7(a) | 3 | theory | What is Fermi Dirac distribution? |
| 2 | 21-22 7(d) | 5 | numerical repeats 20-21 2(d) | At what energy level above Fermi level (in unit of kT) the probability of finding an electron is 10%? |
| 3 | 20-21 2(d) | 10 | numerical | Answer the following questions on Fermi level: (i) At what energy level above Fermi level (in unit of $kT$) the probability of finding an electron is 10% ? (ii) Find the electron concentration($n_0$) for Si at room temperature if Fermi level is 0.5 eV below the conduction band. (at room temperature, $N_c = 2.8 \times 10^{19}$ cm$^{-3}$). |
| 4 | 20-21 TT2 8(a) | 1 | theory | Write down the expression for Fermi Dirac distribution. |
| 5 | 20-21 TT2 8(g) | 5 | numerical repeats 20-21 2(d) | At what energy level above Fermi level (in unit of kT) the probability of finding an electron is 10%? |
| 6 | 20-21 TT2 8(h) | 5 | numerical repeats 20-21 2(d) | Find the electron concentration ($n_o$) for Si at room temperature if Fermi level is 0.5 eV below the conduction band. (at room temperature $N_c = 2.8 \times 10^{19}$ cm$^{-3}$) |
| 7 | 19-20 TT1 7(d) | 5 | numerical | (i) Calculate the probability of an electron to be occupied at the lowest energy state of the conduction band of silicon when the Fermi energy is 0.3 $eV$ below the conduction band at room temperature. (ii) Calculate the thermal electron concentration $n_0$. (iii) What is the thermal hole concentration $p_0$? (Room temperature $N_c$ is $2.8 \times 10^{19}$ $cm^{-3}$, $n_i$ is $1.5 \times 10^{10}$ $cm^{-3}$) |
| 8 | 18-19 1(c) | 5 | theory | At absolute zero temperature, what is the maximum energy of an electron in a semiconductor material? |
| 9 | 18-19 2(b) | 6 | theory | What is Fermi energy? $kT$ is equal to how many $eV$ at 300$k$? |
| **Density of states** | | | | |
| 10 | 20-21 1(c) | 5 | numerical | Determine the number of quantum states(cm$^{-3}$) in silicon between $E_c$ and $E_c$+$kT$ at $T = 300$ K. |
| 11 | 18-19 7(a) | 2 | theory | Define the density of states and carrier concentration. |
| **Effective mass** | | | | |
| 12 | 18-19 7(c) | 3 | theory | What is electron-effective mass? Two possible conduction bands in the E versus k diagram are given in Figure 1. State which band will result in the heavier electron effective mass;state why. |
| **Energy bands** | | | | |
| 13 | 18-19 7(d) | 3 | theory | Using the necessary theorem explains how band structure is formed of a particular solid-state material. |


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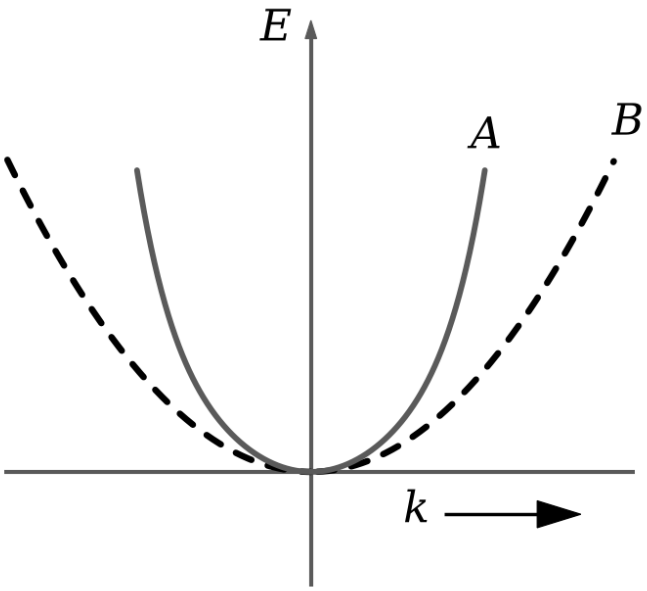



**Figure 1:** 18-19 7(c): Figure 1: Two possible conduction bands A and B in the E versus k diagram (redrawn from the paper)

## 4.3. Semiconductor in equilibrium:

130 marks, 32 questions, in 4 of 4 papers. Book: Neamen Ch. 4.

Most often theory (16 of 32). Subtopics: Temperature dependence; Carrier concentrations; Fermi level; Compensation; Carrier distribution; Ionization; Dopants; Degenerate.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Temperature dependence (12) | **Intrinsic concentration** | $n_i{}^2 = N_c N_v \exp\left(-\frac{E_g}{kT}\right)$ | 18-19 1(a), 18-19 7(b) |
| | **Majority carriers** | $p_0 = \frac{N_a - N_d}{2} + \sqrt{\left(\frac{N_a - N_d}{2}\right)^2 + n_i^2}$ (and $n_0$ with $N_d - N_a$) | 18-19 1(a), 18-19 1(b), 18-19 8(b) |
| | Electrons left on donors | $\frac{n_d}{n_d + n_0} = \frac{1}{1 + \frac{N_c}{2N_d}\exp\left[-\frac{E_c - E_d}{kT}\right]}$ | 20-21 1(d), 20-21 6(b), 20-21 7(a), 19-20 2(c), 18-19 1(a) |
| | $n_i$ at another temperature | $n_i(T) = n_i(300)\left(\frac{T}{300}\right)^{3/2}\exp\left[-\frac{E_g}{2k}\left(\frac{1}{T} - \frac{1}{300}\right)\right]$ | 19-20 2(c) |
| Carrier concentrations (56) | **Electron concentration** | $n_0 = \int_{E_c}^{\infty} g_c(E) f_F(E)\, dE = N_c \exp\left[-\frac{E_c - E_F}{kT}\right]$ | 18-19 7(b) |
| | **Mass-action law** | $n_0 p_0 = n_i^2$ | 18-19 1(b) |
| | **Mass-action law** | $n_0 p_0 = n_i^2 = N_c N_v \exp\left(-\frac{E_g}{kT}\right)$ | 21-22 7(c), 20-21 1(a), 20-21 7(h), 19-20 2(a), 19-20 7(a) |


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| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| | **Band gap from $n_i$** | $E_g = kT \ln\left(\frac{N_c N_v}{n_i^2}\right),\ n_i^2 = n_0(n_0 - N_d)$ | 19-20 7(c) |
| | **Charge neutrality** | $n_0 + N_a^- = p_0 + N_d^+$ | 21-22 7(f), 20-21 3(a), 20-21 8(f), 19-20 2(a), 19-20 7(a) |
| | **Electrons** | $n_0 = \frac{N_d - N_a}{2} + \sqrt{\left(\frac{N_d - N_a}{2}\right)^2 + n_i^2}$ | 20-21 3(a), 20-21 7(g), 20-21 8(f), 19-20 2(a), 19-20 7(a) |
| | **Holes** | $p_0 = \frac{N_a - N_d}{2} + \sqrt{\left(\frac{N_a - N_d}{2}\right)^2 + n_i^2}$ | 21-22 7(f), 20-21 3(a), 20-21 8(f), 19-20 2(a), 19-20 7(a) |
| | Carriers per unit energy | $n_0 = \int_{E_c}^{\infty} g_c(E) f_F(E)\, dE,\ p_0 = \int_{-\infty}^{E_v} g_v(E)[1 - f_F(E)]\, dE$ | 20-21 7(f) |
| | **From the intrinsic level** | $n_0 = n_i \exp\left(\frac{E_F - E_{Fi}}{kT}\right),\ p_0 = n_i \exp\left(-\frac{E_F - E_{Fi}}{kT}\right)$ | 20-21 7(f) |
| | Intrinsic at high $T$ | $n_0 = \frac{N_d}{2} + \sqrt{\left(\frac{N_d}{2}\right)^2 + n_i^2(T)},\ n_i^2 \propto T^3 e^{-E_g/kT}$ | 20-21 7(e) |
| | Integral used | $\int_0^{\infty} \eta^{1/2} e^{-\eta}\, d\eta = \frac{\sqrt{\pi}}{2}$ | 21-22 7(c) |
| | **From the intrinsic level** | $n_0 = n_i \exp\left[\frac{E_F - E_{Fi}}{kT}\right],\ p_0 = n_i \exp\left[-\frac{E_F - E_{Fi}}{kT}\right]$ | 21-22 7(c) |
| | Fermi level, n-type | $E_F - E_{Fi} = kT \ln\frac{n_0}{n_i},\ E_c - E_F = kT \ln\frac{N_c}{n_0}$ | 21-22 7(c) |
| | **Fermi level, p-type** | $E_{Fi} - E_F = kT \ln\frac{p_0}{n_i},\ E_F - E_v = kT \ln\frac{N_v}{p_0}$ | 21-22 7(c), 21-22 7(f) |
| Fermi level (15) | Hole concentration | $p_0 = N_v \exp\left[-\frac{E_F - E_v}{kT}\right]$ | 21-22 7(c), 20-21 1(a), 20-21 7(f), 19-20 3(a), 18-19 8(b) |
| | **Fermi level from the band edge** | $E_F - E_v = kT \ln\frac{N_v}{p_0}$ | 18-19 8(b) |
| | Fermi level from the intrinsic level | $E_{Fi} - E_F = kT \ln\frac{p_0}{n_i}$ | 18-19 8(b) |
| | **Fermi level** | $E_F - E_{Fi} = kT \ln\frac{n_0}{n_i},\ E_{Fi} - E_F = kT \ln\frac{p_0}{n_i}$ | 19-20 7(b) |
| | Fermi level from the band edges | $E_c - E_F = kT \ln\frac{N_c}{n_0},\ E_F - E_v = kT \ln\frac{N_v}{p_0}$ | 19-20 7(b) |

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| | **Fermi level vs doping** | $E_F - E_{Fi} = kT \ln \frac{N_d}{n_i}$ (n-type), $E_{Fi} - E_F = kT \ln \frac{N_a}{n_i}$ (p-type) | 20-21 1(a) |
| | Fermi level from $E_v$ (p-type) | $E_F - E_v = kT \ln \frac{N_v}{N_a}$ | 20-21 8(c) |
| Compensation (19) | Amphoteric split | $N_d = x_{Ga} N_{Si}$, $N_a = x_{As} N_{Si}$ | 21-22 7(f) |
| | Minority carriers | $n_0 = \frac{n_i^2}{p_0}$ | 21-22 7(f) |
| Carrier distribution (8) | **Fermi-Dirac function** | $f_F(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{kT}\right)}$ | 20-21 1(a), 19-20 3(a) |
| | **Density of states** | $g_c(E) = \frac{4\pi(2m_n^*)^{3/2}}{h^3}\sqrt{E - E_c}$ | 21-22 7(c), 19-20 3(a) |
| | Carrier distributions | $n(E) = g_c(E) f_F(E)$, $p(E) = g_v(E)[1 - f_F(E)]$ | 21-22 7(c), 19-20 3(a) |
| | **Electron concentration** | $n_0 = N_c \exp\left[-\frac{E_c - E_F}{kT}\right]$ | 21-22 7(c), 21-22 7(e), 20-21 7(f), 19-20 3(a) |
| Ionization (17) | $N_c$ at another temperature | $N_c(T) = N_c(300)\left(\frac{T}{300}\right)^{3/2}$ | 20-21 1(d) |
| | Fermi level from $E_c$ | $E_c - E_F = kT \ln \frac{N_c}{n_0}$ | 20-21 1(d) |
| | Donor occupation | $f_d = \frac{1}{1 + \frac{1}{2}\exp\left(\frac{E_d - E_F}{kT}\right)}$ | 20-21 1(d), 20-21 7(b) |
| | Acceptors still neutral | $\frac{p_a}{p_a + p_0} = \frac{1}{1 + \frac{N_v}{4N_a}\exp\left[-\frac{E_a - E_v}{kT}\right]}$ | 20-21 7(a) |
| | Freeze-out estimate | $n_0 \approx \sqrt{\frac{N_c N_d}{2}}\exp\left[-\frac{E_c - E_d}{2kT}\right]$ | 20-21 1(d) |
| | Bohr orbit radius | $\frac{r_n}{a_0} = n^2 \epsilon_r \frac{m_0}{m^*}$ | 21-22 7(b) |
| | **Donor energy (hydrogen model)** | $E = -\frac{m^* e^4}{2(n\hbar)^2(4\pi\epsilon)^2}$ | 21-22 7(b) |
| | **Ionization energy** | $E_{ion} = 13.6\text{ eV} \times \frac{m^*/m_0}{\epsilon_r^2}$ | 21-22 7(b) |
| | Donor and acceptor | donor: $E_c - E_d$; acceptor: $E_a - E_v$ | 21-22 7(b), 21-22 7(f) |
| Dopants (2) | Si as a dopant in GaAs | Si on a Ga site: donor; Si on an As site: acceptor | 20-21 7(d), 20-21 8(d) |
| Degenerate (1) | Degenerate limit | $E_F$ inside a band when $n_0 > N_c$ (or $p_0 > N_v$) | 20-21 8(e) |


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## Symbols:

$n_d$ = electrons still in donor states ↩

$n_i$ = intrinsic carrier concentration ↩

$N_a^-$, $N_d^+$ = ionized acceptors and donors ($cm^{-3}$) ↩

$E_{Fi}$ = intrinsic Fermi level (near midgap) ↩

$N_v$ = effective density of states in the valence band ↩

$f_F$ = probability that a state at energy $E$ holds an electron ↩

$E_F$ = Fermi level ↩

$N_c$ = effective density of states in the conduction band ↩

$E_d$ = donor level ↩

$m^*$ = conductivity effective mass ↩

## Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Temperature dependence** | | | | |
| 1 | 19-20 2(c) | 5 | theory repeats 18-19 1(a) | Using proper graph, explain how carrier concentration varies with temperature. |
| 2 | 18-19 1(a) | 7 | theory | Explain the temperature dependance *[sic: should be "dependence"]* of carrier concentration of an extrinsic semiconductor with the help of graph. |
| **Carrier concentrations** | | | | |
| 3 | 21-22 7(c) | 5 | derivation repeats 20-21 7(f) | Derive the equations of equilibrium distribution of electron and holes for extrinsic semiconductor material. |
| 4 | 21-22 7(e) | 0 | numerical repeats 20-21 2(d) | Find the electron concentration ($n_o$) for Si at room temperature if Fermi level is 0.5 eV below the conduction band. (at room temperature $N_c$ =2.8 × $10^{19}$ $cm^{-3}$) |
| 5 | 20-21 3(a) | 5 | derivation repeats 19-20 7(a) | Starting from charge neutrality condition $n_0 + N_a^- = p_0 + N_d^+$, derive the expressions for $n_0$ and $p_0$ in terms of $N_a$, $N_d$, and $n_i$. |
| 6 | 20-21 TT1 7(e) | 1 | theory | True or false: At high temperature extrinsic semiconductor stars *[sic: should be "starts"]* behaving like intrinsic semiconductor? |
| 7 | 20-21 TT1 7(f) | 5 | derivation repeats 20-21 3(a) | Derive the equations of equilibrium distribution of electron and holes for extrinsic semiconductor material. |
| 8 | 20-21 TT1 7(g) | 5 | numerical | What will be the majority carrier concentration of a n-type semiconductor at 300K with $N_d = 10^6$ $cm^{-3}$ *[sic: should be* $10^{16}$ $cm^{-3}$*]* and carrier concentration of $n_i = 1.5 \times 10^{10}$ $cm^{-3}$. |
| 9 | 20-21 TT1 7(h) | 5 | numerical repeats 20-21 2(d) | What will be the minority carrier concentration. |
| 10 | 20-21 TT2 8(f) | 5 | derivation repeats 20-21 3(a) | Starting from charge neutrality condition $n_o + N^{a-}$ *[sic: should be* $N_a^-$*]* $= p_o + N_d^+$, derive expressions for $n_o$ and $p_o$ in terms of $N_a$, $N_d$, and $n_i$. |
| 11 | 19-20 2(a) | 6 | derivation | Starting from charge neutrality condition, Find thermal equilibrium carrier concentration at compensated semiconductor in terms of $N_a$, $N_d$ and $n_i$ |
| 12 | 19-20 TT1 7(a) | 5 | derivation repeats 19-20 2(a) | Starting from charge neutrality condition, derive expressions for $n_0$ and $p_0$ in terms of $N_a$, $N_d$, and $n_i$. |
| 13 | 19-20 TT1 7(c) | 5 | numerical | A special semiconductor material is to be "designed". The semiconductor is to be n-type and doped with $1 \times 10^{15} cm^{-3}$ donor atoms. Assume complete ionization and assume $N_a = 0$. The effective density of states functions are given by $N_c = N_v = 1.5 \times 10^{19} cm^{-3}$ and are independent of temperature. A particular semiconductor device fabricated with this material requires the electron concentration to be no greater than $1.01 \times 10^{15} cm^{-3}$ at $T = 400\ K$. What is the minimum value of the bandgap energy? |
| 14 | 18-19 1(b) | 6 | numerical | A germanium sample is doped with $10^{15}$ boron atoms per $cm^3$. Find the electron concentration. Intrinsic carrier concentration of germanium is $2.5 \times 10^{13}/cm^3$ at $300k$. |


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| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| 15 | 18-19 7(b) | 3 | theory | Explain with the help of equations and graphs- why intrinsic carrier concentration ($n_i$) for a typical semiconductor is high ( $10^6$-$10^9$/cc) at 300K even though the Fermi Dirac distribution is low at that temperature. |
| **Fermi level** | | | | |
| 16 | 20-21 1(a) | 5 | theory | Sketch the position of Fermi level in Si at 300 K as a function of doping concentration. |
| 17 | 20-21 TT2 8(c) | 1 | diagram | Sketch $E_c$, $E_v$, $E_{Fi}$ and $E_F$ for a p type extrinsic semiconductor. |
| 18 | 19-20 TT1 7(b) | 5 | theory | With proper equation and graph, explain variation of Fermi energy level in extrinsic semiconductor. |
| 19 | 18-19 TT2 8(b) | 4 | numerical | Determine the position of the Fermi level with respect to the valence band energy in p type GaAs at T = 300 K. The doping concentrations are [unclear: $N_a$] $= 5 \times 10^{19}$ *[sic: should be* $5 \times 10^{16}$*]* $\text{cm}^{-3}$ and $N_d = 10^{15}$ *[sic: should be* $4 \times 10^{15}$*]* $\text{cm}^{-3}$. |
| **Compensation** | | | | |
| 20 | 21-22 7(f) | 7 | numerical | Silicon atoms, at a concentration of $7 \times 10^{15}$ $\text{cm}^{-3}$, are added to gallium arsenide. Assume that the silicon atoms act as fully ionized dopant atoms and that 5 percent of the concentration added replaces gallium atoms and 95 percent replaces arsenic atoms. Let T = 300 K and GaAs has an intrinsic concentration of 1.8x$10^6$ cm $^{-3}$. (i) Determine the donor and acceptor concentration. (ii) Is the material n-type or p-type? (iii) Calculate the electron and hole concentration. (iv) Determine the position of the Fermi level with respect to $E_{Fi}$. |
| 21 | 20-21 2(a) | 5 | theory repeats 19-20 2(d) | Draw energy band diagram of a compensated semiconductor. |
| 22 | 20-21 TT2 8(b) | 1 | theory | What is compensated semiconductor? |
| 23 | 19-20 2(d) | 6 | theory | Sketch and briefly describe energy band diagram of compensated semiconductor. |
| **Carrier distribution** | | | | |
| 24 | 19-20 3(a) | 8 | theory | Sketch and explain density of states functions, Fermi distribution, and carrier concentration distribution curves of n-type material in a single graph. |
| **Ionization** | | | | |
| 25 | 21-22 7(b) | 0 | theory | Explain Ionization Energy. |
| 26 | 20-21 1(d) | 10 | numerical | A Si sample is doped with $N_d = 1 \times 10^{15}$ $\text{cm}^{-3}$ of As atoms. Calculate the electron and hole concentrations and the position of Fermi level position (w.r.t. $E_c/E_v$) at 300 K, assuming full ionization. Justify the assumption by calculating the probability of doner *[sic: should be donor]* states being occupied by electrons and therefore not ionized (assume, $E_C - E_D = 50$ meV). Repeat the above calculations for $T = 30$ K and comment on the values and assumptions. [$N_c = 2.8 \times 10^{19}$ $\text{cm}^{-3}$, for Si]. |
| 27 | 20-21 6(b) | 5 | derivation | Briefly explain the terms complete ionization and freeze out. Can we achieve complete ionization in room temperature? Prove your answer. |
| 28 | 20-21 TT1 7(a) | 1 | theory | What fraction of donor or acceptor states are ionized at room temperature? |
| 29 | 20-21 TT1 7(b) | 1 | theory | What is the probability function of electron occupying the donor state?? |
| **Dopants** | | | | |
| 30 | 20-21 TT1 7(d) | 1 | theory | Can Si be used as a donor? How? |
| 31 | 20-21 TT2 8(d) | 1 | theory repeats 20-21 7(d) | Can Si be used as a donor? How? |


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| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Degenerate** | | | | |
| 32 | 20-21 TT2 8(e) | 1 | theory | What is degenerated semiconductor? |

## 4.4. Carrier transport:

75 marks, 12 questions, in 3 of 4 papers. Book: Neamen Ch. 5.

Most often numerical (5 of 12). Subtopics: Hall effect; Einstein relation; Diffusion; Mobility; Thermal velocity.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Hall effect (25) | **Hall voltage** | $V_H = \frac{I_x B_z}{epd}$; $p = \frac{I_x B_z}{edV_H}$, $n = -\frac{I_x B_z}{edV_H}$ | 18-19 1(d) |
| | Hall mobility | $\mu_p = \frac{I_x L}{epV_x W d}$ | 18-19 1(d) |
| | **Hall voltage** | $V_H = E_H W$, $V_H = -\frac{I_x B_z}{ned}$ | 19-20 2(b) |
| | Hall concentration and mobility | $n = -\frac{I_x B_z}{edV_H}$, $\mu_n = \frac{I_x L}{enV_x W d}$ | 19-20 2(b) |
| | **Hall voltage** | $V_H = \frac{I_x B_z}{epd}$ (p-type), $V_H = -\frac{I_x B_z}{end}$ (n-type) | 20-21 5(d) |
| | **Carrier density, Hall** | $n = -\frac{I_x B_z}{edV_H}$ | 20-21 5(d) |
| | **Hall mobility** | $\mu_n = \frac{I_x L}{enV_x W d}$ | 20-21 5(d) |
| Einstein relation (11) | **Einstein relation** | $\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = \frac{kT}{e}$ | 20-21 2(b), 18-19 2(c) |
| Diffusion (7) | **Diffusion current** | $J_{n,dif} = eD_n\frac{dn}{dx}$, $J_{p,dif} = -eD_p\frac{dp}{dx}$ | 18-19 5(a) |
| | Total current | $J = en\mu_n E + ep\mu_p E + eD_n\frac{dn}{dx} - eD_p\frac{dp}{dx}$ | 18-19 5(a) |
| Mobility (25) | **Mass-action law** | $n_0 p_0 = n_i^2$ | 18-19 8(c) |
| | **Drift and conductivity** | $\sigma = e(\mu_n n + \mu_p p)$, $\rho = \frac{1}{\sigma}$ | 18-19 8(c) |
| | **Mobility** | $\mu = \frac{e\tau}{m^*}$, $v_d = \mu E$ | 19-20 4(a) |
| | **Two scattering mechanisms** | $\mu_L \propto T^{-3/2}$, $\mu_I \propto \frac{T^{3/2}}{N_I}$, $\frac{1}{\mu} = \frac{1}{\mu_L} + \frac{1}{\mu_I}$ | 20-21 4(a), 19-20 4(a), 19-20 3(d) |
| | **Conductivity** | $\sigma = e(\mu_n n + \mu_p p)$ | 19-20 3(d) |
| | **Drift velocity** | $v_d = \mu E$ | 20-21 3(c) |
| | **Mobility** | $\mu = \frac{e\tau}{m^*}$ | 20-21 3(c) |


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| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| | Conductivity | $\sigma = e(n\mu_n + p\mu_p)$ | 20-21 3(c) |
| Thermal velocity (7) | Maxwell-Boltzmann speeds | $f(v) = 4\pi \left(\frac{m}{2\pi kT}\right)^{3/2} v^2 \exp\left(-\frac{mv^2}{2kT}\right)$ | 19-20 1(c) |
| | **Mean speed** | $\bar{v} = \sqrt{\frac{8kT}{\pi m^*}}$ | 19-20 1(c) |
| | rms (thermal) speed | $v_{rms} = \sqrt{\frac{3kT}{m^*}}$ | 19-20 1(c) |

## Symbols:

$d$ = thickness along $B_z$

$L$ = length along $x$, $W$ = width along $y$

$\tau$ = mean time between collisions (s) ↩

$\mu_L$, $\mu_I$ = lattice- and impurity-limited mobility ↩

$N_I$ = ionized impurity concentration ↩

$\mu$ = mobility ($\text{cm}^2/\text{V s}$) ↩

$E$ = electric field (V/cm) ↩

$m^*$ = conductivity effective mass ↩

$m$ = effective mass of the carrier ↩

## Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Hall effect** | | | | |
| 1 | 20-21 5(d) | 10 | numerical | Current is flowing through an extrinsic semiconductor placed in a magnetic field as shown in Fig. 1. (i) What is the Hall voltage? (ii) What is type of carrier? (iii) What is the carrier concentration? (iv) What is the mobility? |
| 2 | 19-20 2(b) | 8 | numerical | Consider silicon at $T = 300$ K. A Hall effect device is fabricated with the following geometry: $d = 5 \times 10^{-3}$ cm, $W = 5 \times 10^{-2}$ cm and $L = 0.50$ cm. The electrical parameters measured are: $I_x = 0.50$ mA, $V_x = 1.25$ V, and $B_z = 650$ *gauss* $= 6.5 \times 10^{-2}$ *tesla*. The Hall field is $E_H = -16.5$ mV $\text{cm}^{-1}$. Determine: (i) the Hall voltage (ii) the conductivity type (iii) the majority carrier concentration and (iv) the majority carrier mobility |
| 3 | 18-19 1(d) | 7 | derivation | What is the Hall effect? Derive the expression for finding the carrier concentration of a semiconductor from Hall voltage. |
| **Einstein relation** | | | | |
| 4 | 20-21 2(b) | 5 | derivation repeats 18-19 2(c) | Derive the Einstein relation relating the mobility and diffusion constant. Also, discuss the significance of the Einstein relation. |
| 5 | 18-19 2(c) | 6 | derivation | What is Einstein Relation? Derive the expression. |
| **Diffusion** | | | | |
| 6 | 18-19 5(a) | 7 | derivation | Describe diffusion process. Derive the expression for diffusion current density. |
| **Mobility** | | | | |
| 7 | 20-21 3(c) | 5 | theory | Define drift mobility. Explain the mechanism of carrier scattering in semiconductor materials and significance of $\tau$. |
| 8 | 20-21 4(a) | 5 | theory repeats 19-20 4(a) | Explain the effects of temperature on carrier mobility inside a semiconductor. |
| 9 | 19-20 3(d) | 6 | numerical | Consider a compensated n-type silicon at $T = 300$ K with a conductivity of $\rho$ *[sic: should be* $\sigma$*]* $= 16\ (\Omega - \text{cm})^{-1}$ and an acceptor doping concentration of $10^{17}$ $\text{cm}^{-3}$. Determine the doner *[sic: should be donor]* concentration and the electron mobility. |
| 10 | 19-20 4(a) | 6 | theory | With proper graphs, explain how mobility is affected by temperature and doping concentration. |


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| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| 11 | 18-19 TT2 8(c) | 3 | numerical | Silicon at T = 300 K is doped with impurity concentrations of Nd = 5 × 1016 *[sic: should be* $10^{16}$*]* /cm3 and Na = 2 × 1016 *[sic: should be* $10^{16}$*]* /cm3. (a) What are the electron and hole mobilities? (b) Determine the conductivity and resistivity of the [unclear: material]. |
| **Thermal velocity** | | | | |
| 12 | 19-20 1(c) | 7 | numerical | If the free electrons in a semiconductor have effective mass $m_c = 0.1m_o$, where $m_o$ is the mass of the electron in vacuum, what is their average speed at $T = 300K$? If the free holes in the same material have effective mass $m_v = 1.5m_o$, what is their average speed at the same temperature? Assume a Maxwell-Boltzmann distribution of the electron energies. |

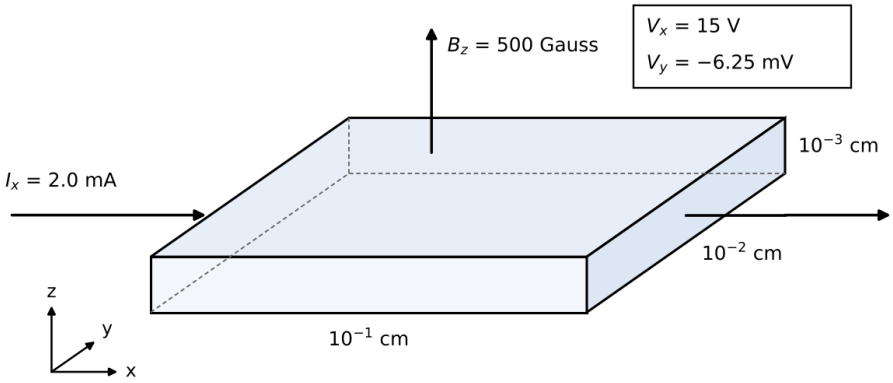



**Figure 2:** 20-21 5(d): Fig. 1: A bar of an extrinsic semiconductor in a magnetic field (redrawn from the paper)

## 4.5. Excess carriers:

78 marks, 13 questions, in 3 of 4 papers. Book: Neamen Ch. 6.

Most often derivation (6 of 13). Subtopics: Steady state; Dielectric relaxation; Ambipolar transport; Time and space dependence; Quasi-Fermi levels; Generation and recombination; Minority carriers; Carrier pulse; Excess carriers.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Steady state (29) | **Diffusion length** | $L_p = \sqrt{D_p \tau_{p0}}$, $\delta p(x) = \delta p_0 e^{-x/L_p}$ | 18-19 5(c) |
| | **Steady state, source at $x = 0$** | $\delta n(x) = \delta n(0) e^{-\lvert x\rvert/L_n}$, $L_n = \sqrt{D_n \tau_{n0}}$ | 19-20 5(a), 19-20 8(a), 19-20 4(d) |
| | **Ambipolar transport** | $D' \frac{\partial^2(\delta n)}{\partial x^2} + \mu' E \frac{\partial(\delta n)}{\partial x} + g' - R' = \frac{\partial(\delta n)}{\partial t}$ | 20-21 3(d) |
| | **Steady-state diffusion** | $\delta n(x) = \delta n(0) \exp\left(-\frac{x}{L_n}\right)$, $L_n = \sqrt{D_n \tau_{n0}}$ | 20-21 3(d) |
| Dielectric relaxation (7) | **Dielectric relaxation** | $\rho(t) = \rho(0) e^{-t/\tau_d}$, $\tau_d = \frac{\epsilon}{\sigma}$ | 19-20 3(c) |
| Ambipolar transport (14) | Current density | $J = en\mu_n E + ep\mu_p E + eD_n \frac{dn}{dx} - eD_p \frac{dp}{dx}$ | 19-20 4(b) |
| | Einstein relation | $\frac{\mu_n}{D_n} = \frac{\mu_p}{D_p} = \frac{e}{kT}$ | 19-20 4(b) |


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| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| | Continuity | $\frac{\partial p}{\partial t} = -\frac{\partial F_p^+}{\partial x} + g_p - \frac{p}{\tau_{pt}}$ | 19-20 4(b), 19-20 8(b) |
| | **Ambipolar transport equation** | $D'\frac{\partial^2(\delta n)}{\partial x^2} + \mu' E\frac{\partial(\delta n)}{\partial x} + g - R = \frac{\partial(\delta n)}{\partial t}$ | 19-20 4(b), 19-20 8(b) |
| | **Ambipolar coefficients** | $D' = \frac{D_n D_p(n+p)}{D_n n + D_p p},\ \mu' = \frac{\mu_n \mu_p (p-n)}{\mu_n n + \mu_p p}$ | 19-20 4(b), 19-20 8(b) |
| Time and space dependence (6) | **Uniform decay and build-up** | $\delta p(t) = \delta p(0) e^{-t/\tau_{p0}},\ \delta p(t) = g'\tau_{p0}(1 - e^{-t/\tau_{p0}})$ | 19-20 4(d) |
| | Pulse | $\delta p(x,t) = \frac{e^{-t/\tau_{p0}}}{(4\pi D_p t)^{1/2}} \exp\left[-\frac{(x-\mu_p E_0 t)^2}{4D_p t}\right]$ | 19-20 4(d) |
| Quasi-Fermi levels (4) | **Quasi-Fermi levels** | $E_{Fn} - E_{Fi} = kT\ln\frac{n_0+\delta n}{n_i},\ E_{Fi} - E_{Fp} = kT\ln\frac{p_0+\delta p}{n_i}$ | 19-20 5(c) |
| Generation and recombination (7) | Direct recombination | $\frac{dn}{dt} = \alpha_r(n_i^2 - np),\ R' = \frac{\delta n}{\tau_{n0}}$ | 19-20 8(c) |
| | SRH recombination | $R = \frac{C_n C_p N_t (np - n_i^2)}{C_n(n+n') + C_p(p+p')}$ | 19-20 8(c) |
| Minority carriers (5) | **Diffusion current** | $J_n = eD_n\frac{dn}{dx},\ J_p = -eD_p\frac{dp}{dx}$ | 20-21 1(b) |
| | **Poisson's equation** | $\frac{d^2\phi}{dx^2} = -\frac{\rho(x)}{\epsilon_s} = -\frac{dE}{dx}$ | 20-21 1(b) |
| Carrier pulse (5) | n-type, low injection | $D_p\frac{\partial^2(\delta p)}{\partial x^2} - \mu_p E_0\frac{\partial(\delta p)}{\partial x} - \frac{\delta p}{\tau_{p0}} = \frac{\partial(\delta p)}{\partial t}$ | 20-21 5(a) |
| | **Moving pulse** | $\delta p(x,t) = \frac{N_0}{\sqrt{4\pi D_p t}} \exp\left(-\frac{t}{\tau_{p0}}\right) \exp\left[-\frac{(x-\mu_p E_0 t)^2}{4D_p t}\right]$ | 20-21 5(a) |
| Excess carriers (1) | Total and excess carriers | $n = n_0 + \delta n,\ p = p_0 + \delta p$ | 20-21 7(c) |

## Symbols:

$L_n$ = minority-electron diffusion length ↩

$F_p^+$ = hole flux ↩

$\tau_{p0}$ = minority carrier lifetime ↩

$N_t$ = trap density ↩

$D_n$, $D_p$ = diffusion coefficients ($\text{cm}^2/\text{s}$) ↩

$\phi$ = potential (V) ↩

$\rho$ = charge density ($\text{C}/\text{cm}^3$) ↩

$N_0$ = pairs created per unit area at $x = 0$, $t = 0$ ↩

$n_0$, $p_0$ = equilibrium values ↩

$\delta n$, $\delta p$ = excess ↩

## Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Steady state** | | | | |
| 1 | 20-21 3(d) | 10 | numerical | For the following questions: (i) Solve the ambipolar transport equation for a semiconductor with $N_a = 5 \times 10^{16}$ cm$^{-3}$ for $x > 0$ at steady state with zero electric field where excess charges are created only at $x = 0$. ($\delta n_0 = 5 \times 10^{15}$ cm$^{-3}$). (ii) Find the distance where the excess charge will drop to $10 \times 10^{13}$ cm$^{-3}$. (Assume $\tau_{n0} = 7 \times 10^{-7}$ s, $\tau_{p0} = 5 \times 10^{-7}$ s, $D_n = 30$ cm$^2$ s$^{-1}$, $D_p = 25$ cm$^2$ s$^{-1}$) (iii) Plot n and p as function of x in log scale. |
| 2 | 19-20 5(a) | 7 | derivation | Consider a p-type semiconductor that is homogeneous and infinite in extent. Assume a zero applied electric field. For a one-dimensional crystal, assume that excess carriers are being generated at $x = 0$ only. The excess carriers being generated at $x = 0$ will being *[sic: should be will begin]* diffusing in both the $-x$ and $+x$ directions. Calculate the steady-state excess carrier concentration as a function of $x$. |
| 3 | 19-20 TT2 8(a) | 6 | derivation repeats 19-20 5(a) | Consider a p-type semiconductor that is homogeneous and infinite in extent. Assume a zero applied electric field. For a one-dimensional crystal, assume that excess carriers are being generated at $x = 0$ only. The excess carriers being generated at $x = 0$ will begin diffusing in both the $-x$ and $+x$ directions. Calculate the steady-state excess carrier concentration as a function of $x$. |
| 4 | 18-19 5(c) | 6 | derivation | Show that diffusion length is the average length a carrier diffuses before recombination. |
| **Dielectric relaxation** | | | | |
| 5 | 19-20 3(c) | 7 | derivation | Explain dielectric relaxation time. Find the equation for dielectric relaxation. |
| **Ambipolar transport** | | | | |
| 6 | 19-20 4(b) | 7 | derivation | Define ambipolar transport. Find ambipolar transport equation. Hence show that diffuse and drift behavior of excess majority carrier is determined by excess minority carrier. |
| 7 | 19-20 TT2 8(b) | 7 | derivation repeats 19-20 4(b) | Define ambipolar transport. Find ambipolar transport equation. Hence show that diffuse and drift behavior of excess majority carrier is determined by excess minority carrier. |
| **Time and space dependence** | | | | |
| 8 | 19-20 4(d) | 6 | theory | With proper graphs, explain time and spatial dependence of excess carrier concentration. |
| **Quasi-Fermi levels** | | | | |
| 9 | 19-20 5(c) | 4 | numerical | Consider an n-type semiconductor at $T = 300$ K with carrier concentrations of $n_o = 10^{15}$ cm$^{-3}$, $n_i = 10^{10}$ cm$^{-3}$ and $p_o = 10^5$ cm$^{-3}$. In non-equilibrium, assume that the excess carrier concentrations are $\delta n = \delta p = 10^{-13}$ *[sic: should be* $10^{13}$*]* cm$^{-3}$. Calculate the quasi-Fermi energy levels. |
| **Generation and recombination** | | | | |
| 10 | 19-20 TT2 8(c) | 7 | theory | Briefly describe different types of generation and recombination processes. Use appropriate diagram if necessary. |
| **Minority carriers** | | | | |
| 11 | 20-21 1(b) | 5 | theory | How do you define a minority carrier? Why does it diffuse? What is Poisson's equation and why is it important in PN junction? |
| **Carrier pulse** | | | | |
| 12 | 20-21 5(a) | 5 | theory | Write the ambipolar equation for $g' = 0$ for n type semiconductor with constant electric field $E_0$. At time $t = 0$, a number of electron-hole pair is generated at $x = 0$. Sketch excess electron concentration as a function of distance at different time. |
| **Excess carriers** | | | | |
| 13 | 20-21 TT1 7(c) | 1 | theory | What are n, p, $n_o$ and $p_o$? |


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## 4.6. PN junction:

72 marks, 12 questions, in 3 of 4 papers. Book: Neamen Ch. 7, 8.

Most often numerical (5 of 12). Subtopics: Switching transient; Junction capacitance; Ideal diode current; Depletion width; Built-in potential; Band diagram; Biased junction.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Switching transient (7) | **Storage time** | $\text{erf}\sqrt{\frac{t_s}{\tau_{p0}}} = \frac{I_F}{I_F + I_R}$ | 18-19 2(a) |
| Junction capacitance (6) | **Built-in potential** | $V_{bi} = \frac{kT}{e} \ln\left(\frac{N_a N_d}{n_i^2}\right)$ | 19-20 3(b), 19-20 4(c), 19-20 5(d), 18-19 2(d), 18-19 8(a) |
| | **Total width** | $W = \left\{\frac{2\epsilon_s(V_{bi}+V_R)}{e}\left[\frac{N_a+N_d}{N_a N_d}\right]\right\}^{1/2}$ | 19-20 4(c), 19-20 5(d), 18-19 2(d), 18-19 8(a) |
| | **Junction capacitance** | $C' = \frac{\epsilon_s}{W} = \left\{\frac{e\epsilon_s N_a N_d}{2(V_{bi}+V_R)(N_a+N_d)}\right\}^{1/2}$ | 18-19 2(d) |
| Ideal diode current (15) | **Ideal diode** | $J = \left[\frac{eD_p p_{n0}}{L_p} + \frac{eD_n n_{p0}}{L_n}\right]\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right]$ | 19-20 5(b), 18-19 4(a) |
| | **Law of the junction** | $p_n(x_n) = p_{n0}e^{eV_a/kT}$, $n_p(-x_p) = n_{p0}e^{eV_a/kT}$ | 19-20 5(b) |
| | Current components | $J_p(x) = J_p(x_n)e^{(x_n-x)/L_p}$, $J_n(x) = J - J_p(x)$ | 19-20 5(b) |
| Depletion width (25) | **Charge balance** | $N_a x_p = N_d x_n$ | 20-21 2(c), 18-19 8(a) |
| | **Space charge widths** | $x_n = \left[\frac{2\epsilon_s V_{bi}}{e}\frac{N_a}{N_d}\frac{1}{N_a+N_d}\right]^{1/2}$, $x_p = \left[\frac{2\epsilon_s V_{bi}}{e}\frac{N_d}{N_a}\frac{1}{N_a+N_d}\right]^{1/2}$ | 18-19 8(a) |
| | $n_i$ at another temperature | $n_i(T) = n_i(300)\left(\frac{T}{300}\right)^{3/2}\exp\left[-\frac{E_g}{2k}\left(\frac{1}{T}-\frac{1}{300}\right)\right]$ | 19-20 4(c) |
| | Split of $W$ | $x_n = W\frac{N_a}{N_a+N_d}$, $x_p = W\frac{N_d}{N_a+N_d}$ | 20-21 2(c), 19-20 4(c), 19-20 5(d) |
| | **Peak field** | $\lvert E_{max}\rvert = \frac{eN_d x_n}{\epsilon_s} = \frac{2(V_{bi}+V_R)}{W}$ | 19-20 4(c), 19-20 5(d) |
| | **Depletion width** | $W = x_n + x_p = \left\{\frac{2\epsilon_s V_{bi}}{e}\left[\frac{N_a+N_d}{N_a N_d}\right]\right\}^{1/2}$ | 20-21 2(c) |
| | **Peak field** | $E_{max} = \frac{eN_d x_n}{\epsilon_s} = \frac{eN_a x_p}{\epsilon_s} = \frac{2V_{bi}}{W}$ | 20-21 6(d) |
| Built-in potential (4) | Current density | $J = en\mu_n E + ep\mu_p E + eD_n\frac{dn}{dx} - eD_p\frac{dp}{dx}$ | 19-20 3(b) |
| | Barrier and minority carriers | $n_{p0} = n_{n0}\exp\left(-\frac{eV_{bi}}{kT}\right)$ | 19-20 3(b), 19-20 5(b) |


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| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Band diagram (5) | **Fermi potentials** | $\phi_{Fp} = V_t \ln \frac{N_a}{n_i}$, $\lvert\phi_{Fn}\rvert = V_t \ln \frac{N_d}{n_i}$ | 20-21 3(b) |
| | **Built-in potential** | $V_{bi} = \lvert\phi_{Fn}\rvert + \lvert\phi_{Fp}\rvert = V_t \ln \frac{N_a N_d}{n_i^2}$ | 20-21 3(b), 20-21 6(d) |
| Biased junction (10) | **Law of the junction** | $p_n(x_n) = p_{n0} \exp\left(\frac{eV_a}{kT}\right)$, $n_p(-x_p) = n_{p0} \exp\left(\frac{eV_a}{kT}\right)$ | 20-21 4(d) |
| | Majority carriers at the edge | $n_n = N_d + \delta p_n$ | 20-21 4(d) |

## Symbols:

$x_n$, $x_p$ = depletion widths on the n and p sides ↩

## Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Switching transient** | | | | |
| 1 | 18-19 2(a) | 7 | theory | What is storage delay time? What is the impact of reverse recovery time of a junction in switching application? |
| **Junction capacitance** | | | | |
| 2 | 18-19 2(d) | 6 | theory | Wat *[sic: should be "What"]* is the depletion capacitance of a $PN$ junction? Explain its variation with reverse bias voltage. |
| **Ideal diode current** | | | | |
| 3 | 19-20 5(b) | 8 | derivation repeats 18-19 4(a) | Find ideal pn junction current and sketch ideal electron and hole current components. |
| 4 | 18-19 4(a) | 7 | diagram | Sketch the ideal electron and hole current components through a $pn$ junction under forward bias. |
| **Depletion width** | | | | |
| 5 | 20-21 2(c) | 0 | derivation | Using the expression for $V_{bi}$, find the width of the depletion region. |
| 6 | 20-21 6(d) | 10 | numerical repeats 18-19 8(a) | A silicon PN junction at $T = 300$ K with zero applied bias has doping concentration of $N_d = 5 \times 10^{16}$ cm$^{-3}$ and $N_a = 5 \times 10^{15}$ cm$^{-3}$. Determine built-in potential, space charge width and electric field at metallurgical region ($\epsilon_t = 11.7$ *[sic: should be* $\epsilon_r = 11.7$*]*) |
| 7 | 19-20 4(c) | 6 | numerical | For a pn junction with $N_a = 1.5 \times 10^{18}$cm$^{-3}$, $N_d = 1.5 \times 10^{16}$cm$^{-3}$ and $n_i = 2 \times 10^{10}$cm$^{-3}$ (i) calculate the built-in potential barrier $V_{bi}$ (ii) calculate the depletion region with *[sic: should be width]* and $E_{max}$ (Assume $\epsilon_s = 11.7\epsilon_o$) and (iii) what will be the width at liquid nitrogen temperature (80K). |
| 8 | 19-20 5(d) | 6 | numerical | An abrupt silicon pn junction has dopant concentrations of $N_a = 2 \times 10^{16}$ cm$^{-3}$ and $N_d = 2 \times 10^{15}$ cm$^{-3}$ at $T = 300$ K. Calculate: (i) $V_{bi}$ (ii) $W$ at $V_R = 0$ and $V_R = 8$ V and (iii) the maximum electric field in the space charge region at $V_R = 0$ and $V_R = 8$ V |
| 9 | 18-19 TT2 8(a) | 3 | numerical | A silicon pn junction at T = 300 K with zero applied bias has doping concentrations of $N_d = 5 \times 10^{16}$ cm$^{-3}$ and $N_a = 5 \times 10^{20}$ *[sic: should be* $5 \times 10^{15}$*]* cm$^{-3}$. Determine $x_n$, $x_p$ and $W$. |
| **Built-in potential** | | | | |
| 10 | 19-20 3(b) | 4 | theory | Explain how barrier potential maintains equilibrium p, n regions. |
| **Band diagram** | | | | |
| 11 | 20-21 3(b) | 5 | theory | Sketch Energy-band diagram of a PN junction in thermal equilibrium and show P region, N region, $E_C$, $E_V$, $E_F$, $E_{Fi}$, $e\phi_{Fn}$, $e\phi_{Fp}$ and $eV_{bi}$. |

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Biased junction** | | | | |
| 12 | 20-21 4(d) | 10 | numerical | Consider a PN junction at room temperature with $n_i = 1.5 \times 10^{10}$ cm$^{-3}$, assume doping in N side is $1 \times 10^6$ cm$^{-3}$ *[sic: should be* $1 \times 10^{16}$ *cm$^{-3}$]*. What will be the minority carrier hole concentrations at the edge in N side if forward bias and reverse bias of 0.7 V is applied? Do the same for majority carrier electron concentration in N side. |

## 4.7. Tunnel diode:

11 marks, 2 questions, in 2 of 4 papers. Book: Neamen Ch. 8, 15.

Most often theory (2 of 2). Subtopics: Operation.

### Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Operation** | | | | |
| 1 | 20-21 5(c) | 5 | theory repeats 18-19 4(e) | Explain the working principle of Tunnel diode. Draw its characteristics curve. |
| 2 | 18-19 4(e) | 6 | theory | Explain the working of Tunnel diode. Draw its characteristics curve. |

## 4.8. Metal-semiconductor contacts:

16 marks, 3 questions, in 2 of 4 papers. Book: Neamen Ch. 9.

Most often theory (3 of 3). Subtopics: Schottky and ohmic contacts.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Schottky and ohmic contacts (16) | **Schottky barrier** | $\phi_{B0} = \phi_m - \chi,\ V_{bi} = \phi_{B0} - \phi_n,\ \phi_n = \frac{kT}{e} \ln \frac{N_c}{N_d}$ | 18-19 6(d) |
| | Thermionic emission | $J = A^* T^2 \exp\left(-\frac{e\phi_{B0}}{kT}\right)\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right]$ | 18-19 6(d), 18-19 4(c) |
| | Schottky barrier | $\phi_{B0} = \phi_m - \chi,\ V_{bi} = \phi_{B0} - \phi_n$ | 19-20 1(d) |
| | Thermionic emission | $J = J_{sT}\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right],\ J_{sT} = A^* T^2 \exp\left(-\frac{e\phi_{Bn}}{kT}\right)$ | 19-20 1(d) |

### Symbols:

$\phi_m$ = metal work function ↩ $\chi$ = electron affinity (4.01 V for Si), $e\phi_n = E_c - E_F$ ↩ $A^*$ = effective Richardson constant ↩

### Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Schottky and ohmic contacts** | | | | |
| 1 | 19-20 1(d) | 6 | theory repeats 18-19 4(c) | Explain Schottky barrier. How it differs from ohmic contact? Explain. |
| 2 | 18-19 4(c) | 4 | theory | Differentiate between ohmic and rectifying contacts. |
| 3 | 18-19 6(d) | 6 | theory | With suitable energy band diagrams, explain Schottky contact. |

## 4.9. MOS structure:

54 marks, 9 questions, in 3 of 4 papers. Book: Neamen Ch. 10.

Most often theory (6 of 9). Subtopics: Band diagrams; Threshold voltage; Body effect; C-V characteristics; MOSFET advantages; Double gate; Inversion.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Band diagrams (13) | **Fermi potential** | $\phi_{fp} = V_t \ln \frac{N_a}{n_i}$ | 20-21 4(b), 19-20 6(a), 18-19 3(a), 18-19 3(b) |
| Threshold voltage (18) | **Threshold voltage** | $V_{TN} = \frac{\lvert Q'_{SD}(\max)\rvert - Q'_{ss}}{C_{ox}} + \phi_{ms} + 2\phi_{fp},\ x_{dT} = \left(\frac{4\epsilon_s \phi_{fp}}{eN_a}\right)^{1/2}$ | 18-19 3(b) |
| | **Maximum depletion** | $x_{dT} = \left(\frac{4\epsilon_s \phi_{fp}}{eN_a}\right)^{1/2},\ \lvert Q'_{SD}(max)\rvert = eN_a x_{dT}$ | 19-20 6(a) |
| | **Oxide field limit** | $\epsilon_{ox} E_{ox} = \lvert Q'_{SD}\rvert + \lvert Q'_I\rvert$ | 19-20 6(a) |
| | **Threshold voltage** | $V_T = V_{FB} + 2\phi_{fp} + \frac{\lvert Q'_{SD}(max)\rvert}{C_{ox}},\ C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$ | 19-20 6(a) |
| | Gate voltage above threshold | $V_G = V_T + \frac{\lvert Q'_I\rvert}{C_{ox}} = V_{FB} + 2\phi_{fp} + E_{ox} t_{ox}$ | 19-20 6(a) |
| Body effect (5) | **Body effect** | $\Delta V_T = \gamma\left[\sqrt{2\phi_{fp} + V_{SB}} - \sqrt{2\phi_{fp}}\right],\ \gamma = \frac{\sqrt{2e\epsilon_s N_a}}{C_{ox}}$ | 18-19 3(c) |
| C-V characteristics (5) | **MOS capacitance** | $\frac{1}{C'} = \frac{1}{C_{ox}} + \frac{1}{C'_{SD}},\ C'_{min} = \frac{\epsilon_{ox}}{t_{ox} + (\epsilon_{ox}/\epsilon_s) x_{dT}}$ | 18-19 5(b) |
| Inversion (5) | **Threshold inversion** | $\phi_s = 2\phi_{fp}$ | 20-21 6(c) |
| | Maximum depletion width | $x_{dT} = \left[\frac{4\epsilon_s \phi_{fp}}{eN_a}\right]^{1/2}$ | 20-21 6(c) |


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| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|

## Symbols:

$\phi_s$ = surface potential (band bending) ↩

## Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Band diagrams** | | | | |
| 1 | 20-21 4(b) | 5 | theory repeats 18-19 3(a) | Draw the energy band diagram of a Metal-Oxide-Semiconductor structure for the three modes of operation of MOSFET and thus explain how the channel is formed. |
| 2 | 18-19 3(a) | 8 | diagram | With brief description sketch the energy band diagram of a MOS structure at different bias voltage. (Use no bias condition, positive bias voltage and negative bias voltage). |
| **Threshold voltage** | | | | |
| 3 | 19-20 6(a) | 12 | numerical | The body of a nanoscale N-channel MOSFET is doped at $N_A = 5^{18}$ *[sic: should be* $5 \times 10^{18}$*]* $\mathrm{cm}^{-3}$. (i) Determine the maximum inversion-layer charge per unit area ($Q_I$) that can be reached so that the electric field in the oxide remains below $E_{ox} = 0.6$ V/nm. Neglect any interface-trap and gate-oxide charge. (ii) If the channel length and the channel width are $L = 50$ nm and $W = 500$ nm, respectively, determine how many electrons are creating this inversion layer. (iii) Calculate the threshold voltage of a MOSFET having gate-oxide thickness $t_{ox} = 1$ nm and flat-band voltage $V_{FB} = 1.0$ V. (iv) What gate voltage is needed to form the maximum inversion-layer charge? |
| 4 | 18-19 3(b) | 6 | theory | What are the parameters that affect the magnitude of threshold voltage in a MOSFET? |
| **Body effect** | | | | |
| 5 | 18-19 3(c) | 5 | theory | What is body effect of MOSFET? |
| **C-V characteristics** | | | | |
| 6 | 18-19 5(b) | 5 | diagram | Draw the ideal low-frequency capacitance versus gate voltage curve of a MOS capacitor with p-type substrate and indicate different regions. |
| **MOSFET advantages** | | | | |
| 7 | 19-20 6(c) | 3 | theory | Mention some advantages of using MOSFET. |
| **Double gate** | | | | |
| 8 | 20-21 6(a) | 5 | theory | Draw the schematic of a double-gate MOS structure. What are the benefits of such structure over single-gate MOS? |
| **Inversion** | | | | |
| 9 | 20-21 6(c) | 5 | theory | Explain production of inverse layer at oxide and semiconductor layer with proper sketches of energy band diagram. |


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## 4.10. Bipolar transistor:

47 marks, 9 questions, in 3 of 4 papers. Book: Neamen Ch. 12.

Most often theory (7 of 9). Subtopics: Amplification; Base current; Base width modulation; Emitter efficiency; Current crowding; Operating modes; Minority distribution.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Amplification (6) | **Collector current** | $i_C = I_S \exp\left(\frac{v_{BE}}{V_t}\right)$, $A_v = -g_m R_C = -\frac{I_C R_C}{V_t}$ | 18-19 3(d) |
| Base current (3) | Base current | $I_B \approx I_{pE} + I_{RB} + I_R - I_{CBO}$ | 18-19 4(b) |
| Base width modulation (5) | **Early effect** | $\frac{dI_C}{dV_{CE}} = \frac{I_C}{V_{CE}+V_A}$, $J_C \approx \frac{eD_B n_{B0}}{x_B} e^{eV_{BE}/kT}$ | 18-19 4(d) |
| Emitter efficiency (10) | **Injection efficiency** | $\gamma \approx \frac{1}{1+\frac{N_B}{N_E}\frac{D_E}{D_B}\frac{x_B}{x_E}}$ | 18-19 6(b) |
| | **Emitter injection efficiency** | $\gamma = \frac{J_{nE}}{J_{nE}+J_{pE}} \approx \frac{1}{1+\frac{N_B}{N_E}\frac{D_E}{D_B}\frac{x_B}{x_E}}$ | 19-20 6(d) |
| Current crowding (7) | Current crowding | $\Delta V = \frac{I_B}{2}R$, injection $\propto e^{eV_{BE}/kT}$ | 18-19 6(c) |
| Operating modes (11) | **BJT modes** | active: $V_{BE}>0$, $V_{BC}<0$; saturation: both $>0$; cutoff: both $<0$; inverse: $V_{BE}<0$, $V_{BC}>0$ | 19-20 6(b) |
| | **Equilibrium minority carriers** | $p_{E0} = \frac{n_i^2}{N_E}$, $n_{B0} = \frac{n_i^2}{N_B}$, $p_{C0} = \frac{n_i^2}{N_C}$ | 20-21 4(c) |
| | **Four modes** | active: $V_{BE}>0$, $V_{CB}>0$; saturation: $V_{BE}>0$, $V_{CB}<0$; cutoff: $V_{BE}<0$, $V_{CB}>0$; inverse: $V_{BE}<0$, $V_{CB}<0$ | 20-21 4(c) |
| Minority distribution (5) | **Base edges** | $n_B(0) = n_{B0}\exp\left(\frac{V_{BE}}{V_t}\right)$, $n_B(x_B) = n_{B0}\exp\left(\frac{V_{BC}}{V_t}\right)$ | 20-21 5(b) |
| | Emitter and collector edges | $p_E(0) = p_{E0}\exp\left(\frac{V_{BE}}{V_t}\right)$, $p_C(0) = p_{C0}\exp\left(\frac{V_{BC}}{V_t}\right)$ | 20-21 5(b) |


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| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|

### Symbols:

$N_E$, $N_B$, $N_C$ = emitter, base, collector doping ↩ $x_B$ = neutral base width ↩

### Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Amplification** | | | | |
| 1 | 18-19 3(d) | 6 | theory | Explain the amplification action of a transistor. |
| **Base current** | | | | |
| 2 | 18-19 4(b) | 3 | theory | What are the factors which cause base current in a transistor? |
| **Base width modulation** | | | | |
| 3 | 18-19 4(d) | 5 | theory | What is base width modulation? |
| **Emitter efficiency** | | | | |
| 4 | 19-20 6(d) | 4 | theory | Does the doping level in the base of a BJT affect the emitter efficiency $\gamma_E$? If so, why? |
| 5 | 18-19 6(b) | 6 | numerical | Design the ratio of emitter doping to base doping in order to achieve an emitter injection efficiency factor of $\gamma = 0.9967$. Assume a *npn* bipolar transistor, and also consider $D_E = D_B$, $L_E = L_B$ and $x_E = X_B$. |
| **Current crowding** | | | | |
| 6 | 18-19 6(c) | 7 | theory | What is current crowding? What is the impact of current crowding in high power devices? |
| **Operating modes** | | | | |
| 7 | 20-21 4(c) | 5 | theory | Show the four operating modes of a bipolar transistor as a function of $V_{BE}$ and $V_{CB}$. Draw the minority carrier distribution of the four modes of a NPN transistor with proper labelling of all the quantities. |
| 8 | 19-20 6(b) | 6 | numerical | Determine the mode of operation of an NPN BJT with $\beta \approx 450$, if it is known that: (i) $V_{BE} = 0.7V$, $V_{CE} = 5.2V$ (ii) $V_{BE} = 0.7V$, $V_{CE} = 0.2V$ (iii) $V_{BE} = 0.8V$, $V_{BC} = 0.3V$ (iv) $V_{BE} = -0.8V$, $V_{BC} = 0.7V$ (v) $I_C = 455mA$, $I_B = 1mA$ (vi) $I_C = 455mA$, $I_E = 502mA$ |
| **Minority distribution** | | | | |
| 9 | 20-21 5(b) | 5 | theory | For NPN transistor, plot minority carrier concentration in base, emitter, and collector as a function of space. What are the minority carrier concentrations at the boundaries of base, emitter, and collector? |

## 4.11. JFET:

13 marks, 2 questions, in 1 of 4 papers. Book: Neamen Ch. 13.

Most often numerical (2 of 2). Subtopics: Pinch-off; Frequency response.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Pinch-off (7) | **Pinch-off voltage** | $V_{p0} = \frac{ea^2 N_d}{2\epsilon_s}$, $V_p = V_{bi} - V_{p0}$ | 18-19 5(d) |
| Frequency response (6) | **Cutoff frequency** | $f_T = \frac{g_m}{2\pi C_G} = \frac{e\mu_n N_d a^2}{2\pi\epsilon_s L^2}$ | 18-19 6(a) |


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### Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Pinch-off** | | | | |
| 1 | 18-19 5(d) | 7 | numerical | A silicon n-channel JFET at $T = 300K$ has a gate doping concentration of $N_a = 10^{18} cm^{-3}$ and a channel doping concentration of $N_d = 2 \times 10^{16} cm^{-3}$. Determine the metallurgical channel thickness, $a$, such that the pinchoff voltage is $V_p = -2.50\ V$. |
| **Frequency response** | | | | |
| 2 | 18-19 6(a) | 6 | numerical | The parameters of an n-channel silicon JFET are $\mu_n = 1000\ cm^2/V - s$, $N_d = 5 \times 10^{15} cm^{-3}$, $a = 0.50\ \mu m$, and $L = 2\ \mu m$. Determine the cutoff frequency. |

# 5. Theory points:

What a full answer covers, for every theory question, in book order. The link opens the full answer.

| No. | Paper | Marks | Subtopic | Points to cover |
|---|---|---|---|---|
| **Crystal structure** | | | | |
| 1 | 19-20 1(a) | 6 | Classes of solids | Amorphous: order only within a few atomic distances, no grains (glass, a-Si). Polycrystalline: high order inside grains of different orientation, separated by grain boundaries (polysilicon gates). Single crystal: one lattice through the whole volume, no grain boundaries, best electrical properties (Si wafers). |
| **Quantum basics** | | | | |
| 2 | 21-22 7(a) | 3 | Fermi probability | $f_F(E) = 1/[1 + e^{(E-E_F)/kT}]$ is the probability that a state at $E$ holds an electron (Pauli: one per state, particles not told apart). Step at 0 K; 1/2 at $E_F$ for any $T$, rounded over a few $kT$; $1 - f_F$ is the hole probability, symmetric about $E_F$; Boltzmann form $e^{-(E-E_F)/kT}$ for $E - E_F > 3kT$. |
| 3 | 20-21 TT2 8(a) | 1 | Fermi probability | $f_F(E) = 1/[1 + \exp((E - E_F)/kT)]$: probability that a state at $E$ is occupied; 1/2 at $E_F$, a step at 0 K; $1 - f_F$ is the probability of a hole. |
| 4 | 18-19 1(c) | 5 | Fermi probability | At 0 K $f_F$ is a step: every state below $E_F$ full, every state above empty. In a semiconductor $E_F$ lies in the gap, so the valence band is completely full and the conduction band empty: the highest electron energy is $E_v$, the top of the valence band (in a metal it would be $E_F$). |
| 5 | 18-19 2(b) | 6 | Fermi probability | Fermi energy $E_F$: the energy at which a state is occupied with probability 1/2; at 0 K all states below it are full; its position in the gap sets $n_0$ and $p_0$. $kT$ at 300 K: $(1.38 \times 10^{-23})(300)/(1.6 \times 10^{-19}) = 0.0259$ eV $\approx 1/40$ eV. |
| 6 | 18-19 7(a) | 2 | Density of states | Density of states $g(E)$: allowed states per unit energy per unit volume, $g_c = 4\pi(2m_n^*)^{3/2}\sqrt{E - E_c}/h^3$. Carrier concentration: free carriers per unit volume, $n_0 = \int g_c f_F\, dE = N_c e^{-(E_c - E_F)/kT}$. |
| 7 | 18-19 7(c) | 3 | Effective mass | Effective mass: the mass that makes Newton's law work for an electron in the crystal, including the lattice forces: $1/m^* = (1/\hbar^2) d^2E/dk^2$. Band B is flatter (smaller curvature), so it gives the heavier effective mass (about 2.4 times band A from the figure). |
| 8 | 18-19 7(d) | 3 | Energy bands | Isolated atoms have discrete levels; brought together, wave functions overlap and Pauli forces each level to split into $N$ levels, forming bands. Bloch's theorem $\psi = u(x)e^{jkx}$ in the periodic Kronig-Penney potential gives $P' \sin \alpha a/\alpha a + \cos \alpha a = \cos ka$: allowed where the left side lies within $\pm 1$, so allowed bands separated by forbidden gaps. |

| No. | Paper | Marks | Subtopic | Points to cover |
|---|---|---|---|---|
| **Semiconductor in equilibrium** | | | | |
| 9 | 21-22 7(b) | 0 | Ionization | Energy to free a donor's fifth electron ($E_c - E_d$) or to fill an acceptor's missing bond ($E_a - E_v$), leaving a fixed ion and a free carrier. Hydrogen model: $E_{ion} = 13.6(m^*/m_0)/\epsilon_r^2 = 25.8$ meV in Si; measured 0.045 eV (P, B), 0.05 eV (As); close to $kT$, so dopants are fully ionized at 300 K. |
| 10 | 20-21 1(a) | 5 | Fermi level | $E_F - E_{Fi} = kT\ln(N_d/n_i)$ in n-type, $E_{Fi} - E_F = kT\ln(N_a/n_i)$ in p-type; on a log doping axis both are straight lines of 0.06 eV per decade (0.23, 0.35, 0.47, 0.53 eV at $10^{14}$ to $10^{19}$ cm$^{-3}$); $E_F$ moves towards $E_c$ with donors and $E_v$ with acceptors, reaching the band edge at degeneracy. |
| 11 | 20-21 2(a) | 5 | Compensation | Both donors and acceptors in one region; donor electrons fall into the acceptor states, so donors are ionized (+) and acceptors filled (−). For $N_d > N_a$: n-type, $n_0 \approx N_d - N_a$, $E_F$ above $E_{Fi}$; for $N_a > N_d$: p-type; for $N_a = N_d$: behaves intrinsic. |
| 12 | 20-21 TT1 7(a) | 1 | Ionization | Essentially all (complete ionization) because $E_c - E_d = 0.045$ eV $< 2kT$ and $N_c \gg N_d$; fraction still on donors $1/[1 + (N_c/2N_d)e^{-(E_c-E_d)/kT}] = 0.40$ % for P at $10^{16}$ cm$^{-3}$ (99.6 % ionized); B acceptors at $10^{16}$ cm$^{-3}$ about 97.9 % ionized ($g = 4$). |
| 13 | 20-21 TT1 7(b) | 1 | Ionization | $f_d = 1/[1 + \frac{1}{2}\exp((E_d - E_F)/kT)]$, $n_d = N_d - N_d^+$; the $\frac{1}{2}$ ($1/g$, $g = 2$) comes from the two spin states of a donor level that holds only one electron; $f_d = 2/3$ at $E_d = E_F$. |
| 14 | 20-21 TT1 7(d) | 1 | Dopants | Yes, in a III-V compound such as GaAs: Si (4 valence electrons) on a Ga site (3) has one left over and is a donor (0.0058 eV); on an As site (5) it is an acceptor, so Si is amphoteric, mostly a donor in GaAs; in Si it is the host atom. |
| 15 | 20-21 TT1 7(e) | 1 | Carrier concentrations | True: $n_i \propto e^{-E_g/2kT}$ rises steeply and exceeds the doping at high $T$, so $n_0 \approx p_0 \approx n_i$ and $E_F \to E_{Fi}$; for $N_d = 10^{15}$ cm$^{-3}$, $n_i = N_d$ at about 565 K. |
| 16 | 20-21 TT2 8(b) | 1 | Compensation | Both donors and acceptors in the same region; donor electrons fill the acceptor states, so only the net doping gives carriers: $n_0 \approx N_d - N_a$ if $N_d > N_a$, $p_0 \approx N_a - N_d$ if $N_a > N_d$, intrinsic-like if equal. |
| 17 | 20-21 TT2 8(d) | 1 | Dopants | Yes, in GaAs: Si on a Ga site has one valence electron more than Ga and gives it to the conduction band (0.0058 eV); on an As site it is an acceptor (amphoteric), mostly a donor; in Si it is the host. |
| 18 | 20-21 TT2 8(e) | 1 | Degenerate | So heavily doped ($\gtrsim 10^{18}$ cm$^{-3}$ in Si) that the impurity band overlaps the band edge and $E_F$ lies inside the conduction band ($n_0 > N_c$) or the valence band ($p_0 > N_v$); Boltzmann and $n_0 p_0 = n_i^2$ fail; used in tunnel diodes and ohmic contacts. |
| 19 | 19-20 2(c) | 5 | Temperature dependence | Freeze-out at low $T$ (donors not ionized, $n_0 < N_d$), extrinsic plateau ($n_0 = N_d$, about 100 to 500 K for $10^{15}$ cm$^{-3}$), intrinsic region at high $T$ where $n_i \propto T^{3/2}e^{-E_g/2kT}$ overtakes $N_d$ and $n_0 \to n_i$. |
| 20 | 19-20 2(d) | 6 | Compensation | Donor electrons drop into acceptor levels, so all dopants are ionized; $N_d > N_a$: n-type, $n_0 \approx N_d - N_a$, $E_F$ above $E_{Fi}$; $N_a > N_d$: p-type, $p_0 \approx N_a - N_d$, $E_F$ below $E_{Fi}$; $N_d = N_a$: intrinsic-like; mobility set by $N_d + N_a$. |
| 21 | 19-20 3(a) | 8 | Carrier distribution | $g_c \propto \sqrt{E - E_c}$, $g_v \propto \sqrt{E_v - E}$, zero in the gap; $f_F$ with $E_F$ above $E_{Fi}$ for n-type; $n(E) = g_c f_F$ and $p(E) = g_v(1 - f_F)$ peak $kT/2$ inside each band; areas $n_0 \gg p_0$. |
| 22 | 19-20 TT1 7(b) | 5 | Fermi level | $E_F - E_{Fi} = kT\ln(n_0/n_i)$, $E_{Fi} - E_F = kT\ln(p_0/n_i)$, or $E_c - E_F = kT\ln(N_c/N_d)$, $E_F - E_v = kT\ln(N_v/N_a)$: $E_F$ moves 0.06 eV per decade of doping towards the majority band edge, and back towards $E_{Fi}$ as $T$ rises and $n_i$ grows; freeze-out at very low $T$. |


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| No. | Paper | Marks | Subtopic | Points to cover |
|---|---|---|---|---|
| 23 | 18-19 1(a) | 7 | Temperature dependence | Neutrality $n_0 = N_d^+ + p_0$ with partial ionization. Freeze-out at low $T$ (electrons stay on donors, $n_0 \to 0$ at 0 K); extrinsic plateau $n_0 = N_d$ (about 100 to 500 K for $10^{15}$ cm$^{-3}$ in Si); intrinsic region where $n_i = \sqrt{N_c N_v} e^{-E_g/2kT}$ overtakes $N_d$. $E_F$ moves from near $E_c$ down to $E_{Fi}$ as $T$ rises. |
| 24 | 18-19 7(b) | 3 | Carrier concentrations | $n_0 = \int g_c f_F \, dE$: $f_F$ at $E_c$ is tiny ($\sim 10^{-10}$ in Si with $E_F$ at midgap) but there are very many states ($N_c = 2.8 \times 10^{19}$ cm$^{-3}$); the product gives $n_i = \sqrt{N_c N_v} e^{-E_g/2kT} \sim 10^{10}$ cm$^{-3}$. |
| **Carrier transport** | | | | |
| 25 | 20-21 3(c) | 5 | Mobility | Drift mobility $\mu = \lvert v_d \rvert / E = e\tau/m^*$. Scattering by lattice vibrations ($\mu_L \propto T^{-3/2}$) and ionized impurities ($\mu_I \propto T^{3/2}/N_I$), combined by $1/\mu = 1/\mu_L + 1/\mu_I$. The mean free time $\tau$ (about $10^{-13}$ s) sets the mobility and so the conductivity. |
| 26 | 20-21 4(a) | 5 | Mobility | Lattice scattering grows with $T$, so $\mu_L \propto T^{-3/2}$ falls; impurity scattering weakens as carriers get faster, so $\mu_I \propto T^{3/2}/N_I$ rises; by Matthiessen's rule impurity scattering dominates at low $T$ (mobility rises), lattice at high $T$ (mobility falls); heavier doping lowers the curve. |
| 27 | 19-20 4(a) | 6 | Mobility | Lattice scattering $\mu_L \propto T^{-3/2}$, impurity scattering $\mu_I \propto T^{3/2}/N_I$, $1/\mu = 1/\mu_L + 1/\mu_I$: $\mu$ rises with $T$ at low $T$, falls at high $T$; flat at low doping (about 1350 and 480 cm$^2$/V s in Si), falls steeply above $10^{16}$ cm$^{-3}$ of total ionized impurity. |
| **Excess carriers** | | | | |
| 28 | 20-21 1(b) | 5 | Minority carriers | Minority carrier: the less numerous carrier ($p_{n0} = n_i^2/N_d$). It diffuses because random thermal motion in a concentration gradient gives a net flow from high to low; injected minority carriers recombine as they spread, so their current is diffusion current. Poisson: $d^2\phi/dx^2 = -\rho/\epsilon_s$ links the space charge to the field and potential, giving $E_{max}$, $V_{bi}$ and $W$ of a junction. |
| 29 | 20-21 5(a) | 5 | Carrier pulse | For n-type at low injection the ambipolar equation uses $D_p$, $\mu_p$, $\tau_{p0}$: $D_p \, \partial^2 \delta p/\partial x^2 - \mu_p E_0 \, \partial \delta p/\partial x - \delta p/\tau_{p0} = \partial \delta p/\partial t$. A pulse at $x = 0$ becomes a Gaussian that spreads, shrinks by recombination and drifts in the direction of $E_0$ at $\mu_p E_0$. |
| 30 | 20-21 TT1 7(c) | 1 | Excess carriers | $n_0$, $p_0$: thermal-equilibrium concentrations, $n_0 p_0 = n_i^2$; $n = n_0 + \delta n$, $p = p_0 + \delta p$: total concentrations including excess carriers from light or injection, may vary with time and position. |
| 31 | 19-20 4(d) | 6 | Time and space dependence | Uniform: $\delta p = \delta p(0) e^{-t/\tau_{p0}}$ after the light stops, $g' \tau_{p0}(1 - e^{-t/\tau_{p0}})$ after it starts; steady injection: $\delta p = \delta p(0) e^{-x/L_p}$, $L_p = \sqrt{D_p \tau_{p0}}$; a pulse is a Gaussian that spreads ($4D_p t$), decays ($e^{-t/\tau_{p0}}$) and drifts at $\mu_p E_0$. |
| 32 | 19-20 TT2 8(c) | 7 | Generation and recombination | Band to band: $h\nu \geq E_g$ or heat makes a pair, an electron falling into a hole emits a photon, rate $\alpha_r(np - n_i^2)$, GaAs. SRH: traps in the gap capture and emit electrons and holes (four processes), $R = C_n C_p N_t (np - n_i^2)/[C_n(n + n') + C_p(p + p')]$, non-radiative, dominant in Si. Auger: energy to a third carrier, heavy doping; inverse is impact ionization. Surface recombination through dangling-bond traps. |
| **PN junction** | | | | |
| 33 | 20-21 3(b) | 5 | Band diagram | $E_F$ flat through the device; on the P side $E_F$ is $e\phi_{Fp} = kT \ln(N_a/n_i)$ below $E_{Fi}$, on the N side $e\lvert\phi_{Fn}\rvert = kT \ln(N_d/n_i)$ above it; the bands are higher on the P side by $eV_{bi}$ and bend only inside the space charge region; $V_{bi} = \lvert\phi_{Fp}\rvert + \lvert\phi_{Fn}\rvert$. |
| 34 | 19-20 3(b) | 4 | Built-in potential | Diffusion uncovers ions, giving a field from n to p and a barrier $V_{bi} = (kT/e) \ln(N_a N_d / n_i^2)$; drift cancels diffusion for each carrier ($J_n = J_p = 0$); only $e^{-eV_{bi}/kT}$ of the majority carriers cross, $n_{p0} = n_{n0} e^{-eV_{bi}/kT}$; $E_F$ flat. |


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| No. | Paper | Marks | Subtopic | Points to cover |
|---|---|---|---|---|
| 35 | 18-19 2(a) | 7 | Switching transient | Forward bias stores excess minority charge $Q_p = I_F\tau_{p0}$ near the junction; on switching to reverse a constant $I_R$ flows while the charge is removed (storage time $t_s$, $\text{erf}\sqrt{t_s/\tau_{p0}} = I_F/(I_F + I_R)$), then the current decays (fall time). Total reverse recovery time limits switching speed; cut it with short lifetime (gold doping), small $I_F$, large $I_R$, Schottky diodes. |
| 36 | 18-19 2(d) | 6 | Junction capacitance | Reverse bias widens the space charge region; a change $dV_R$ uncovers charge $dQ'$ at its edges, so $C' = dQ'/dV_R = \epsilon_s/W = \{e\epsilon_s N_a N_d/[2(V_{bi} + V_R)(N_a + N_d)]\}^{1/2}$. It falls as $1/\sqrt{V_{bi} + V_R}$; $(1/C')^2$ is linear in $V_R$ (intercept $-V_{bi}$, slope gives $N_d$). Varactor use. |
| 37 | 18-19 4(a) | 7 | Ideal diode current | Minority diffusion currents $J_p(x_n)$ and $J_n(-x_p)$ at the space charge edges decay as $e^{-(x-x_n)/L_p}$ into the neutral regions; the majority currents rise to keep the total $J$ constant everywhere; no recombination in the space charge region. |
| **Tunnel diode** | | | | |
| 38 | 20-21 5(c) | 5 | Operation | Both sides degenerately doped, so $E_F$ lies inside the bands and the depletion layer is only 5 to 17 nm thick; electrons tunnel where filled states face empty ones; the current rises to a peak $I_p$, falls to a valley $I_v$ (negative resistance), then rises as normal diode current; reverse bias tunnels freely. |
| 39 | 18-19 4(e) | 6 | Operation | Both sides degenerately doped, very thin junction: electrons tunnel between filled n-side states and empty p-side states. Small forward bias: current rises to a peak; more bias: bands uncross, current falls (negative resistance) to a valley; then normal diffusion current. Reverse bias: large tunnelling current. |
| **Metal-semiconductor contacts** | | | | |
| 40 | 19-20 1(d) | 6 | Schottky and ohmic contacts | Metal on n-type with $\phi_m > \phi_s$: electrons leave the semiconductor, a depletion layer and upward band bending form a barrier $\phi_{B0} = \phi_m - \chi$, $V_{bi} = \phi_{B0} - \phi_n$; current by thermionic emission $J = J_{sT}(e^{eV_a/kT} - 1)$, rectifying, majority carriers, fast. Ohmic contact: $\phi_m < \phi_s$ (accumulation, no barrier) or a heavily doped surface so electrons tunnel; linear I-V, negligible resistance. |
| 41 | 18-19 4(c) | 4 | Schottky and ohmic contacts | Rectifying (Schottky): $\phi_m > \phi_s$ on n-type, barrier $\phi_{B0} = \phi_m - \chi$, current by thermionic emission one way only, I-V like a diode. Ohmic: low or negative barrier ($\phi_m < \phi_s$ on n-type) or heavy doping so electrons tunnel through a thin barrier; linear I-V, small resistance either way. |
| 42 | 18-19 6(d) | 6 | Schottky and ohmic contacts | Metal on n-type with $\phi_m > \phi_s$: electrons flow to the metal until $E_F$ is flat, leaving a depletion region and barrier $\phi_{B0} = \phi_m - \chi$, $V_{bi} = \phi_{B0} - \phi_n$. Forward bias lowers $V_{bi}$: thermionic emission $J = A^*T^2 e^{-e\phi_{B0}/kT}(e^{eV_a/kT} - 1)$; reverse bias raises it. Majority carriers, no storage, fast switching; Ex 9.1: 0.54 V barrier. |
| **MOS structure** | | | | |
| 43 | 20-21 4(b) | 5 | Band diagrams | MOS on p-type Si. Accumulation ($V_G < 0$): bands bend up, holes gather at the surface. Depletion ($0 < V_G < V_T$): bands bend down, acceptor ions uncovered over $x_d$. Inversion ($V_G > V_T$): $E_{Fi}$ at the surface falls below $E_F$, electrons form an n-type channel, $\phi_s = 2\phi_{fp}$ at threshold. |
| 44 | 20-21 6(a) | 5 | Double gate | A thin (undoped) silicon body between two thin gate oxides with a gate on each side, $n^+$ source and drain (FinFET is its practical form). Benefits: better gate control of the channel, much smaller short-channel effects (DIBL, $V_T$ roll-off), near-ideal subthreshold swing, low leakage, more drive current, no doping fluctuation. |


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| No. | Paper | Marks | Subtopic | Points to cover |
|---|---|---|---|---|
| 45 | 20-21 6(c) | 5 | Inversion | Positive gate voltage on a p-type body repels holes and bends the bands down by $e\phi_s$: depletion for $\phi_s < \phi_{fp}$, intrinsic surface at $\phi_s = \phi_{fp}$, n-type inversion layer once $E_{Fi}$ falls below $E_F$; threshold at $\phi_s = 2\phi_{fp}$ ($n_s = N_a$), depletion width at its maximum $x_{dT}$. |
| 46 | 19-20 6(c) | 3 | MOSFET advantages | Insulated gate: very high input impedance, voltage control, almost no drive power; CMOS draws almost no static power; small, simple, dense; majority carrier, no storage delay; mobility falls with temperature, no thermal runaway, easy paralleling; good bidirectional switch. |
| 47 | 18-19 3(a) | 8 | Band diagrams | Ideal MOS on p-type: $V_G = 0$ flat bands (flat band). $V_G < 0$: holes attracted, bands bend up, accumulation. Small $V_G > 0$: holes pushed away, bands bend down, depletion region. Large $V_G > 0$: $E_{Fi}$ crosses $E_F$ at the surface, electron inversion layer; threshold at $\phi_s = 2\phi_{fp}$. |
| 48 | 18-19 3(b) | 6 | Threshold voltage | $V_{TN} = (\lvert Q'_{SD}(\max)\rvert - Q'_{ss})t_{ox}/\epsilon_{ox} + \phi_{ms} + 2\phi_{fp}$: substrate doping (through $\phi_{fp}$ and $Q'_{SD}$), oxide thickness and permittivity, fixed oxide charge $Q'_{ss}$, gate material (work function difference $\phi_{ms}$), temperature, body bias (body effect), ion implant adjustment. |
| 49 | 18-19 3(c) | 5 | Body effect | A reverse source-to-body bias $V_{SB}$ widens the depletion region under the gate, adding bulk charge, so more gate voltage is needed: $\Delta V_T = \gamma(\sqrt{2\phi_{fp} + V_{SB}} - \sqrt{2\phi_{fp}})$, $\gamma = \sqrt{2e\epsilon_s N_a}/C_{ox}$ (body-effect coefficient). Matters in stacked transistors. |
| 50 | 18-19 5(b) | 5 | C-V characteristics | p-type MOS: accumulation ($V_G < V_{FB}$): $C = C_{ox}$; depletion: $C_{ox}$ in series with $C'_{SD} = \epsilon_s/x_d$, falls; minimum $C'_{min}$ at threshold ($x_d = x_{dT}$); low-frequency inversion: inversion charge follows, $C$ returns to $C_{ox}$ (high frequency stays at $C_{min}$). |
| **Bipolar transistor** | | | | |
| 51 | 20-21 4(c) | 5 | Operating modes | Forward active ($V_{BE} > 0$, $V_{CB} > 0$), saturation (both junctions forward), cutoff (both reverse), inverse active; a forward junction raises the minority edge value by $e^{V/V_t}$, a reverse one takes it to 0; draw the base, emitter and collector minority profiles for each mode with the edge values. |
| 52 | 20-21 5(b) | 5 | Minority distribution | Forward active: base electrons fall almost linearly from $n_{B0}e^{V_{BE}/V_t}$ at the emitter edge to about 0 at the collector edge; emitter holes fall from $p_{E0}e^{V_{BE}/V_t}$ to $p_{E0}$; collector holes rise from about 0 to $p_{C0}$. |
| 53 | 19-20 6(d) | 4 | Emitter efficiency | Yes: $\gamma = J_{nE}/(J_{nE} + J_{pE}) \approx 1/[1 + (N_B/N_E)(D_E/D_B)(x_B/x_E)]$; heavier base doping lowers $n_{B0} = n_i^2/N_B$, so fewer electrons are injected into the base relative to holes injected back into the emitter, and $\gamma$ falls; hence $N_E \gg N_B$. |
| 54 | 18-19 3(d) | 6 | Amplification | Forward-biased B-E junction injects electrons that cross the thin base and are swept into the collector; $i_C = I_S e^{v_{BE}/V_t}$, so a small $v_{be}$ makes a large change in $i_C$; through $R_C$ this gives $v_{ce} = -g_m R_C v_{be}$, voltage gain $A_v = -I_C R_C/V_t$; current gain $\beta = i_C/i_B$. |
| 55 | 18-19 4(b) | 3 | Base current | Holes injected from the base into the emitter ($I_{pE}$, emitter injection efficiency); recombination of electrons in the base ($I_{RB}$, base transport factor); recombination in the B-E space charge region ($I_R$); minus the reverse B-C leakage $I_{CBO}$. |
| 56 | 18-19 4(d) | 5 | Base width modulation | Raising $V_{CE}$ widens the B-C space charge region into the base, so the neutral base width $x_B$ shrinks; the minority gradient and $I_C \propto 1/x_B$ rise; output curves slope and meet at $-V_A$ (Early voltage); also called the Early effect; finite output resistance. |
| 57 | 18-19 6(c) | 7 | Current crowding | Base current flows sideways under the emitter through the base resistance; the $I_B r_b$ drop makes $V_{BE}$ largest at the emitter edges, and injection $\propto e^{eV_{BE}/kT}$ crowds to the edges (18 mV doubles it). In power devices: local high current density, hot spots, second breakdown, lower gain; cured with narrow interdigitated emitters and ballast resistors. |


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# 6. Derivations:

Every derivation asked, in book order. Each outline gives the steps; the link opens the full working.

| No. | Paper | Marks | Derivation | Outline |
|---|---|---|---|---|
| **Semiconductor in equilibrium** | | | | |
| 1 | 21-22 7(c) | 5 | Carrier concentrations | $n(E) = g_c f_F$; integrate from $E_c$; Boltzmann $f_F$; put in $g_c \propto \sqrt{E - E_c}$; $\eta = (E - E_c)/kT$; $\int \eta^{1/2} e^{-\eta} = \sqrt{\pi}/2$; $n_0 = N_c e^{-(E_c - E_F)/kT}$; same for holes; add and subtract $E_{Fi}$: $n_0 = n_i e^{(E_F - E_{Fi})/kT}$, $p_0 = n_i e^{-(E_F - E_{Fi})/kT}$; product $n_0 p_0 = n_i^2$. |
| 2 | 20-21 3(a) | 5 | Carrier concentrations | Neutrality $n_0 + N_a = p_0 + N_d$ (complete ionization); put $p_0 = n_i^2/n_0$; multiply by $n_0$: quadratic; quadratic formula; keep the positive root; for $p_0$ put $n_0 = n_i^2/p_0$ instead. |
| 3 | 20-21 6(b) | 5 | Ionization | Donor occupation with the factor $\frac{1}{2}$; non-degenerate: $n_d \approx 2N_d e^{-(E_d - E_F)/kT}$; $n_0 = N_c e^{-(E_c - E_F)/kT}$; the ratio cancels $E_F$; fraction $1/[1 + (N_c/2N_d)e^{-(E_c - E_d)/kT}]$; 0.4 % for P in Si at $10^{16}$ $\mathrm{cm}^{-3}$, 300 K. |
| 4 | 20-21 TT1 7(f) | 5 | Carrier concentrations | $n(E) = g_c f_F$; integrate over the band with the Boltzmann form; $\eta = (E - E_c)/kT$ and $\int_0^\infty \eta^{1/2} e^{-\eta} d\eta = \sqrt{\pi}/2$ give $n_0 = N_c e^{-(E_c - E_F)/kT}$; same with $1 - f_F$ for $p_0 = N_v e^{-(E_F - E_v)/kT}$; add and subtract $E_{Fi}$: $n_0 = n_i e^{(E_F - E_{Fi})/kT}$, $p_0 = n_i e^{-(E_F - E_{Fi})/kT}$; product $n_0 p_0 = n_i^2$. |
| 5 | 20-21 TT2 8(f) | 5 | Carrier concentrations | Neutrality $n_0 + N_a = p_0 + N_d$ (complete ionization); put $p_0 = n_i^2/n_0$; multiply by $n_0$: quadratic; quadratic formula; keep the positive root; for $p_0$ put $n_0 = n_i^2/p_0$ instead. |
| 6 | 19-20 2(a) | 6 | Carrier concentrations | Neutrality $n_0 + N_a = p_0 + N_d$ (complete ionization); put $p_0 = n_i^2/n_0$; multiply by $n_0$: quadratic; quadratic formula; keep the positive root; for $p_0$ put $n_0 = n_i^2/p_0$ instead. |
| 7 | 19-20 TT1 7(a) | 5 | Carrier concentrations | Neutrality $n_0 + N_a = p_0 + N_d$ (complete ionization); put $p_0 = n_i^2/n_0$; multiply by $n_0$: quadratic; quadratic formula; keep the positive root; for $p_0$ put $n_0 = n_i^2/p_0$ instead. |
| **Carrier transport** | | | | |
| 8 | 20-21 2(b) | 5 | Einstein relation | Graded n-type sample in equilibrium: $J_n = 0$; $n = N_c e^{-(E_c - E_F)/kT}$ with $E_F$ flat; chain rule gives $dn/dx = -(n/kT) dE_c/dx$; $dE_c/dx = eE_x$; substitute: $D_n/\mu_n = kT/e$; repeat for holes. |
| 9 | 18-19 1(d) | 7 | Hall effect | Current $I_x$ in field $B_z$: Lorentz force $q\mathbf{v} \times \mathbf{B}$ pushes carriers to $y = 0$; charge builds a Hall field until $eE_H = ev_x B_z$; $V_H = E_H W = v_x W B_z$; $v_x = I_x/(epWd)$; $V_H = I_x B_z/(epd)$, so $p = I_x B_z/(edV_H)$; electrons: $n = -I_x B_z/(edV_H)$; sign of $V_H$ gives the type; mobility $\mu = I_x L/(epV_x Wd)$. |
| 10 | 18-19 2(c) | 6 | Einstein relation | Non-uniformly doped n-type in equilibrium: $J_n = en\mu_n E_x + eD_n\, dn/dx = 0$; $E_x = (1/e) dE_c/dx$; $n = N_c e^{-(E_c - E_F)/kT}$ with flat $E_F$ gives $dn/dx = -(en/kT)E_x$; substitute: $\mu_n = eD_n/kT$, so $D_n/\mu_n = kT/e$; same for holes. |
| 11 | 18-19 5(a) | 7 | Diffusion | Carriers move randomly with $v_{th}$ over mean free path $l$; half of those at $\pm l$ cross $x = 0$: $F = \frac{1}{2} v_{th}[n(-l) - n(+l)]$; Taylor $n(\pm l) = n(0) \pm l\, dn/dx$; $F = -v_{th} l\, dn/dx$; $J_n = -eF = eD_n\, dn/dx$ with $D_n = v_{th} l$; holes $J_p = -eD_p\, dp/dx$; total with drift. |
| **Excess carriers** | | | | |
| 12 | 19-20 3(c) | 7 | Dielectric relaxation | Continuity $\nabla \cdot J = -\partial\rho/\partial t$; Ohm $J = \sigma E$; Poisson $\nabla \cdot E = \rho/\epsilon$; $d\rho/dt + (\sigma/\epsilon)\rho = 0$; $\rho = \rho(0) e^{-t/\tau_d}$, $\tau_d = \epsilon/\sigma$ (0.539 ps for $10^{16}$ $\mathrm{cm}^{-3}$ Si). |
| 13 | 19-20 4(b) | 7 | Ambipolar transport | Continuity for holes and electrons; drift-diffusion fluxes; time-dependent diffusion equations; excess form with $\delta n = \delta p$, $g_n = g_p$, $R_n = R_p$; multiply by $\mu_n n$ and $\mu_p p$ and add so $\partial E/\partial x$ cancels; divide: $D'$, $\mu'$; low injection: $D' = D_n$, $\mu' = \mu_n$ in p-type ($D_p$, $-\mu_p$ in n-type), so the majority carriers follow the minority carriers. |


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| No. | Paper | Marks | Derivation | Outline |
|---|---|---|---|---|
| 14 | 19-20 5(a) | 7 | Steady state | Ambipolar equation with minority electron parameters; $E = 0$, $g' = 0$ for $x \neq 0$, steady state; $\delta n'' - \delta n/L_n^2 = 0$; $Ae^{-x/L_n} + Be^{x/L_n}$; finite at $\pm\infty$; $\delta n = \delta n(0)e^{-\lvert x\rvert/L_n}$. |
| 15 | 19-20 TT2 8(a) | 6 | Steady state | Ambipolar equation with minority electron parameters; $E = 0$, $g' = 0$ for $x \neq 0$, steady state; $\delta n'' - \delta n/L_n^2 = 0$; $Ae^{-x/L_n} + Be^{x/L_n}$; finite at $\pm\infty$; $\delta n = \delta n(0)e^{-\lvert x\rvert/L_n}$. |
| 16 | 19-20 TT2 8(b) | 7 | Ambipolar transport | Continuity for holes and electrons; drift-diffusion fluxes; time-dependent diffusion equations; excess form with $\delta n = \delta p$, $g_n = g_p$, $R_n = R_p$; multiply by $\mu_n n$ and $\mu_p p$ and add so $\partial E/\partial x$ cancels; divide: $D'$, $\mu'$; low injection: $D' = D_n$, $\mu' = \mu_n$ in p-type ($D_p$, $-\mu_p$ in n-type), so the majority carriers follow the minority carriers. |
| 17 | 18-19 5(c) | 6 | Steady state | Steady state, no field: $D_p\, d^2\delta p/dx^2 - \delta p/\tau_{p0} = 0$; $L_p = \sqrt{D_p\tau_{p0}}$; $\delta p = \delta p_0 e^{-x/L_p}$; probability of recombining in $dx$: $P(x)dx = (1/L_p)e^{-x/L_p}dx$; mean $\langle x\rangle = (1/L_p)\int_0^\infty xe^{-x/L_p}dx = L_p$ (integration by parts). |
| **PN junction** | | | | |
| 18 | 20-21 2(c) | 0 | Depletion width | Charge density $-eN_a$, $+eN_d$; integrate Poisson once with $E = 0$ at both edges; continuity gives $N_a x_p = N_d x_n$; integrate again for the potential; $V_{bi} = \phi(x_n)$; substitute $x_p = N_d x_n/N_a$; solve for $x_n$, $x_p$; add for $W$. |
| 19 | 19-20 5(b) | 8 | Ideal diode current | $n_{p0} = n_{n0}e^{-eV_{bi}/kT}$; barrier $V_{bi} - V_a$ gives the law of the junction $p_n(x_n) = p_{n0}e^{eV_a/kT}$, $n_p(-x_p) = n_{p0}e^{eV_a/kT}$; minority diffusion equations; decaying exponentials; diffusion currents at the edges; add: $J = [eD_p p_{n0}/L_p + eD_n n_{p0}/L_n](e^{eV_a/kT} - 1)$; majority currents make up the rest. |

# 7. Numericals:

Every numerical asked, in book order: the method in one line and the final answer, so a worked attempt can be checked. The link opens the full working.

| No. | Paper | Marks | Subtopic | Method | Answer |
|---|---|---|---|---|---|
| **Crystal structure** | | | | | |
| 1 | 19-20 1(b) | 6 | Unit cell | fcc: 8 corner atoms shared by 8 cells, 6 face atoms shared by 2: $8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$; volume density $= 4/a^3$ with $a$ in cm. | (i) 4 atoms per unit cell; (ii) $5.21 \times 10^{22}$ cm$^{-3}$ ($a = 0.425$ nm) |
| **Quantum basics** | | | | | |
| 2 | 21-22 7(d) | 5 | Fermi probability | Set $1/[1 + e^{(E-E_F)/kT}] = 0.1$; $e^{(E-E_F)/kT} = 9$; $E - E_F = kT\ln 9$ (do not use the Boltzmann form). | $E - E_F = 2.197\,kT \approx 2.2\,kT$ (0.0569 eV at 300 K) |
| 3 | 20-21 1(c) | 5 | Density of states | Integrate the density of states $g_c(E) = 4\pi(2m_n^*)^{3/2}\sqrt{E - E_c}/h^3$ from $E_c$ to $E_c + kT$: $N = \frac{4\pi(2m_n^*)^{3/2}}{h^3} \cdot \frac{2}{3}(kT)^{3/2}$. | $N = 2.13 \times 10^{19}$ cm$^{-3}$ |
| 4 | 20-21 2(d) | 10 | Fermi probability | (i) Set $f_F = 1/[1 + e^{(E-E_F)/kT}] = 0.1$ and solve: $E - E_F = kT\ln 9$. (ii) $n_0 = N_c \exp[-(E_c - E_F)/kT]$. | (i) $2.2\,kT$ (0.0569 eV); (ii) $1.16 \times 10^{11}$ cm$^{-3}$ |
| 5 | 20-21 TT2 8(g) | 5 | Fermi probability | Set $f_F = 1/[1 + e^{(E-E_F)/kT}] = 0.1$ and solve: $E - E_F = kT\ln 9$. | $2.197\,kT \approx 2.2\,kT$ (0.0569 eV) |
| 6 | 20-21 TT2 8(h) | 5 | Fermi probability | $n_0 = N_c \exp[-(E_c - E_F)/kT]$ with $E_c - E_F = 0.5$ eV. | $n_0 = 1.16 \times 10^{11}$ cm$^{-3}$ |

| No. | Paper | Marks | Subtopic | Method | Answer |
|---|---|---|---|---|---|
| 7 | 19-20 TT1 7(d) | 5 | Fermi probability | $f_F(E_c) = 1/[1 + e^{(E_c - E_F)/kT}]$; $n_0 = N_c e^{-(E_c - E_F)/kT}$; $p_0 = n_i^2/n_0$. | (i) $9.32 \times 10^{-6}$; (ii) $n_0 = 2.61 \times 10^{14}$ cm$^{-3}$; (iii) $p_0 = 8.62 \times 10^5$ cm$^{-3}$ |
| **Semiconductor in equilibrium** | | | | | |
| 8 | 21-22 7(e) | 0 | Carrier concentrations | $n_0 = N_c \exp[-(E_c - E_F)/kT]$ with $E_c - E_F = 0.5$ eV $= 19.3\,kT$. | $n_0 = 1.16 \times 10^{11}$ cm$^{-3}$ |
| 9 | 21-22 7(f) | 7 | Compensation | Si on Ga sites are donors, on As sites acceptors: $N_d = 0.05N$, $N_a = 0.95N$; $N_a > N_d$ so p-type; $p_0 = (N_a - N_d)/2 + \sqrt{((N_a - N_d)/2)^2 + n_i^2}$, $n_0 = n_i^2/p_0$; $E_{Fi} - E_F = kT \ln(p_0/n_i)$. | $N_d = 3.5 \times 10^{14}$, $N_a = 6.65 \times 10^{15}$ cm$^{-3}$; p-type; $p_0 = 6.3 \times 10^{15}$, $n_0 = 5.14 \times 10^{-4}$ cm$^{-3}$; $E_F$ 0.569 eV below $E_{Fi}$ |
| 10 | 20-21 1(d) | 10 | Ionization | $n_0 = N_d$, $p_0 = n_i^2/n_0$; $E_c - E_F = kT \ln(N_c/n_0)$, $E_F - E_v = E_g - (E_c - E_F)$; fraction on donors $1/[1 + (N_c/2N_d)e^{-(E_c - E_d)/kT}]$. | $n_0 = 10^{15}$, $p_0 = 2.25 \times 10^5$ cm$^{-3}$; $E_c - E_F = 0.2652$ eV, $E_F - E_v = 0.8548$ eV; 99.95 % ionized |
| 11 | 20-21 TT1 7(g) | 5 | Carrier concentrations | Neutrality with $p_0 = n_i^2/n_0$: $n_0 = N_d/2 + \sqrt{(N_d/2)^2 + n_i^2}$; with $N_d \gg n_i$, $n_0 = N_d$ (printed $10^6$ read as $10^{16}$). | $n_0 = 1 \times 10^{16}$ cm$^{-3}$ |
| 12 | 20-21 TT1 7(h) | 5 | Carrier concentrations | Mass-action law $p_0 = n_i^2/n_0$. | $p_0 = 2.25 \times 10^4$ cm$^{-3}$ |
| 13 | 19-20 TT1 7(c) | 5 | Carrier concentrations | Neutrality and mass action: $n_i^2 = n_0(n_0 - N_d)$ at $n_0 = 1.01 \times 10^{15}$; $E_g = kT \ln(N_c N_v/n_i^2)$ with $kT = 0.0259 \times 400/300$. | $n_i = 1.005 \times 10^{14}$ cm$^{-3}$; $E_{g,min} = 0.823$ eV |
| 14 | 18-19 1(b) | 6 | Carrier concentrations | Neutrality $p_0 = n_0 + N_a$ with $n_0 = n_i^2/p_0$: $p_0 = N_a/2 + \sqrt{(N_a/2)^2 + n_i^2}$; then $n_0 = n_i^2/p_0$. | $p_0 = 1.0006 \times 10^{15}$ cm$^{-3}$, $n_0 = 6.25 \times 10^{11}$ cm$^{-3}$ |
| 15 | 18-19 TT2 8(b) | 4 | Fermi level | Compensated p-type: $p_0 = N_a - N_d$; $E_F - E_v = kT \ln(N_v/p_0)$ with $N_v = 7.0 \times 10^{18}$ cm$^{-3}$ for GaAs. | $p_0 = 4.6 \times 10^{16}$ cm$^{-3}$; $E_F - E_v = 0.130$ eV |
| **Carrier transport** | | | | | |
| 16 | 20-21 5(d) | 10 | Hall effect | $V_H = V_y$ from the figure; its sign gives the carrier type; $n = -I_x B_z/(edV_H)$ with $d$ the thickness along $z$; $\mu_n = I_x L/(enV_x Wd)$. | $V_H = -6.25$ mV; n-type (electrons); $n = 1.0 \times 10^{16}$ cm$^{-3}$; $\mu_n = 833$ cm$^2$/V s |
| 17 | 19-20 1(c) | 7 | Thermal velocity | Maxwell-Boltzmann speed distribution; mean speed $\bar{v} = \sqrt{8kT/\pi m^*}$ (rms $\sqrt{3kT/m^*}$ for comparison) with $m^* = 0.1m_0$ and $1.5m_0$. | Electrons $3.40 \times 10^5$ m/s, holes $8.78 \times 10^4$ m/s (rms $3.69 \times 10^5$, $9.53 \times 10^4$ m/s) |
| 18 | 19-20 2(b) | 8 | Hall effect | $V_H = E_H W$; its sign gives the carrier type; $n = -I_x B_z/(edV_H)$ with $d$ in metres; $\mu_n = I_x L/(enV_x Wd)$. | $V_H = -0.825$ mV; n-type; $n = 4.92 \times 10^{15}$ cm$^{-3}$; $\mu_n = 1015$ cm$^2$/V s |
| 19 | 19-20 3(d) | 6 | Mobility | $\sigma = e\mu_n(N_d - N_a)$ with $\mu_n$ read at $N_I = N_d + N_a$ from the mobility curve; trial and error ($\rho$ printed for $\sigma$). | $N_d \approx 3.5 \times 10^{17}$ cm$^{-3}$, $\mu_n \approx 400$ cm$^2$/V s |


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| No. | Paper | Marks | Subtopic | Method | Answer |
|---|---|---|---|---|---|
| 20 | 18-19 TT2 8(c) | 3 | Mobility | Mobility from the total $N_I = N_d + N_a$ (Neamen Fig. 5.3); $n_0 = N_d - N_a$; $\sigma = e(\mu_n n_0 + \mu_p p_0) \approx e\mu_n n_0$; $\rho = 1/\sigma$. | $\mu_n \approx 1000$, $\mu_p \approx 350$ cm$^2$/V s; $\sigma = 4.8$ $(\Omega\text{-cm})^{-1}$, $\rho = 0.208$ $\Omega$-cm |
| **Excess carriers** | | | | | |
| 21 | 20-21 3(d) | 10 | Steady state | Low injection in p-type: ambipolar equation becomes $D_n\,\delta n'' - \delta n/\tau_{n0} = 0$; solution decaying for $x > 0$: $\delta n = \delta n_0 e^{-x/L_n}$, $L_n = \sqrt{D_n \tau_{n0}}$; solve $\delta n(x) = 10^{14}$ for $x$. | $\delta n = 5 \times 10^{15} e^{-x/L_n}$, $L_n = 45.8$ $\mu$m; $10^{14}$ cm$^{-3}$ at 179.3 $\mu$m |
| 22 | 19-20 5(c) | 4 | Quasi-Fermi levels | $E_{Fn} - E_{Fi} = kT \ln[(n_0 + \delta n)/n_i]$, $E_{Fi} - E_{Fp} = kT \ln[(p_0 + \delta p)/n_i]$ with $\delta n = \delta p = 10^{13}$ cm$^{-3}$ (printed $10^{-13}$). | $E_{Fn} - E_{Fi} = 0.2984$ eV, $E_{Fi} - E_{Fp} = 0.179$ eV ($E_F - E_{Fi} = 0.2982$ eV) |
| **PN junction** | | | | | |
| 23 | 20-21 4(d) | 10 | Biased junction | $p_{n0} = n_i^2/N_d$; law of the junction $p_n(x_n) = p_{n0} e^{eV_a/kT}$ with $V_a = +0.7$ and $-0.7$ V; $n_n = N_d + \delta p$ (neutrality); check low injection. | $p_{n0} = 2.25 \times 10^4$; forward: $p_n = 1.23 \times 10^{16}$, $n_n = 2.23 \times 10^{16}$ cm$^{-3}$ (low injection violated); reverse: $p_n \approx 0$, $n_n \approx 10^{16}$ cm$^{-3}$ |
| 24 | 20-21 6(d) | 10 | Depletion width | $V_{bi} = V_t \ln\!\left(N_a N_d / n_i^2\right)$; $W = \{(2\epsilon_s V_{bi}/e)[(N_a + N_d)/(N_a N_d)]\}^{1/2}$; $x_n = W N_a/(N_a + N_d)$; $E_{max} = -eN_d x_n/\epsilon_s$. | $V_{bi} = 0.718$ V; $W = 0.452$ $\mu$m; $E_{max} = -3.18 \times 10^4$ V/cm |
| 25 | 19-20 4(c) | 6 | Depletion width | $V_{bi} = V_t \ln\!\left(N_a N_d / n_i^2\right)$; $W = \{(2\epsilon_s V_{bi}/e)[(N_a + N_d)/(N_a N_d)]\}^{1/2}$; $E_{max} = -eN_d x_n/\epsilon_s$; at 80 K scale $n_i$ by $(T/300)^{3/2} e^{-E_g/2k(1/T - 1/300)}$ and $V_t$ by $T/300$. | $V_{bi} = 0.820$ V, $W = 0.267$ $\mu$m, $E_{max} = -6.13 \times 10^4$ V/cm; 80 K: $V_{bi} = 1.069$ V, $W = 0.305$ $\mu$m |
| 26 | 19-20 5(d) | 6 | Depletion width | $V_{bi} = V_t \ln\!\left(N_a N_d / n_i^2\right)$; $W = \{(2\epsilon_s (V_{bi} + V_R)/e)[(N_a + N_d)/(N_a N_d)]\}^{1/2}$; $\lvert E_{max}\rvert = 2(V_{bi} + V_R)/W$ at $V_R = 0$ and 8 V. | $V_{bi} = 0.671$ V; $W = 0.691$ and 2.48 $\mu$m; $E_{max} = -1.94 \times 10^4$ and $-6.98 \times 10^4$ V/cm |
| 27 | 18-19 TT2 8(a) | 3 | Depletion width | $V_{bi} = (kT/e) \ln\!\left(N_a N_d / n_i^2\right)$; $x_n = [(2\epsilon_s V_{bi}/e)(N_a/N_d)/(N_a + N_d)]^{1/2}$, $x_p$ with $N_d/N_a$; $W = x_n + x_p$; check $N_a x_p = N_d x_n$. | $V_{bi} = 0.718$ V; $x_n = 0.0411$ $\mu$m, $x_p = 0.411$ $\mu$m, $W = 0.452$ $\mu$m |
| **MOS structure** | | | | | |
| 28 | 19-20 6(a) | 12 | Threshold voltage | $\phi_{fp} = V_t \ln(N_A/n_i)$; $x_{dT} = (4\epsilon_s \phi_{fp}/eN_A)^{1/2}$; $\lvert Q'_{SD}\rvert = eN_A x_{dT}$; $\lvert Q'_I\rvert = \epsilon_{ox} E_{ox} - \lvert Q'_{SD}\rvert$; electrons $= \lvert Q'_I\rvert LW/e$; $V_T = V_{FB} + 2\phi_{fp} + \lvert Q'_{SD}\rvert/C_{ox}$; $V_G = V_T + \lvert Q'_I\rvert/C_{ox}$ ($N_A$ printed $5^{18}$, read $5 \times 10^{18}$). | $\lvert Q_I\rvert = 7.73 \times 10^{-7}$ C/cm$^2$; 1208 electrons; $V_T = 2.39$ V; $V_G = 2.62$ V |


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| No. | Paper | Marks | Subtopic | Method | Answer |
|---|---|---|---|---|---|
| **Bipolar transistor** | | | | | |
| 29 | 19-20 6(b) | 6 | Operating modes | $V_{BC} = V_{BE} - V_{CE}$; B-E and B-C forward or reverse decide the mode; from currents compare $I_C/I_B$ with $\beta$ ($I_B = I_E - I_C$). | i forward active; ii saturation; iii saturation (soft); iv inverse active; v forward active; vi saturation ($I_C/I_B = 9.7$) |
| 30 | 18-19 6(b) | 6 | Emitter efficiency | $\gamma = 1/[1 + (N_B/N_E)(D_E/D_B)(L_B/L_E)\tanh(x_B/L_B)/\tanh(x_E/L_E)]$; with equal $D$, $L$ and $x$: $\gamma = 1/(1 + N_B/N_E)$, so $N_E/N_B = 1/(1/\gamma - 1)$. | $N_E/N_B = 302$ |
| **JFET** | | | | | |
| 31 | 18-19 5(d) | 7 | Pinch-off | $V_{bi} = V_t \ln(N_a N_d/n_i^2)$; $V_{p0} = V_{bi} - V_p$; $a = [2\epsilon_s V_{p0}/(eN_d)]^{1/2}$. | $V_{bi} = 0.832$ V, $V_{p0} = 3.33$ V, $a = 0.464$ $\mu$m |
| 32 | 18-19 6(a) | 6 | Frequency response | $f_T = g_m/(2\pi C_G)$ with $g_m(\max) = e\mu_n N_d W a/L$ and $C_G(\min) = \epsilon_s WL/a$: $f_T = e\mu_n N_d a^2/(2\pi\epsilon_s L^2)$. | $f_T = 7.69$ GHz |

# 8. Diagrams to practise:

Every figure an answer needs, whatever the question's type, in book order. Draw each one from memory, then check it against the solution.

| No. | Paper | Kind | Diagram |
|---|---|---|---|
| **Crystal structure** | | | |
| 1 | 19-20 1(a) | Diagram | atoms in amorphous, polycrystalline and single-crystal solids |
| 2 | 19-20 1(b) | Diagram | fcc unit cell with corner and face atoms |
| **Quantum basics** | | | |
| 3 | 21-22 7(a) | Graph | Fermi-Dirac function at 0 K, 300 K and 600 K |
| 4 | 21-22 7(a) | Graph | $f_F$ and $1 - f_F$ symmetric about $E_F$ |
| 5 | 21-22 7(d) | Graph | Fermi-Dirac function with the 10 % point |
| 6 | 20-21 1(c) | Graph | density of states with the shaded strip |
| 7 | 20-21 2(d) | Graph | Fermi-Dirac function with the 10 % point |
| 8 | 20-21 TT2 8(a) | Graph | Fermi-Dirac function at three temperatures |
| 9 | 20-21 TT2 8(g) | Graph | Fermi-Dirac function with the 10 % point |
| 10 | 19-20 TT1 7(d) | Diagram | Si band diagram with $E_F$ 0.3 eV below $E_c$ |
| 11 | 18-19 1(c) | Graph | Fermi-Dirac function at 0 K |
| 12 | 18-19 1(c) | Diagram | full valence band and empty conduction band at 0 K |
| 13 | 18-19 2(b) | Graph | Fermi-Dirac function with $f_F = 1/2$ at $E_F$ |
| 14 | 18-19 7(a) | Graph | density of states in the conduction band of Si |
| 15 | 18-19 7(c) | Graph | E-k diagram of bands A and B (Figure 1) |
| 16 | 18-19 7(c) | Graph | parabolic fits of bands A and B with their curvatures |
| 17 | 18-19 7(d) | Diagram | splitting of atomic levels into bands as atoms come together |
| 18 | 18-19 7(d) | Graph | Kronig-Penney function with allowed and forbidden bands |
| 19 | 18-19 7(d) | Graph | E-k diagram from the Kronig-Penney model |
| **Semiconductor in equilibrium** | | | |
| 20 | 21-22 7(b) | Diagram | donor electron orbiting its ion in the Si lattice |
| 21 | 21-22 7(b) | Diagram | donor and acceptor ionization on the band diagram |


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| No. | Paper | Kind | Diagram |
|---|---|---|---|
| 22 | 21-22 7(c) | Graph | density of states, Fermi function and electron and hole distributions for n-type and p-type |
| 23 | 21-22 7(e) | Diagram | Si band diagram with $E_F$ 0.5 eV below $E_c$ |
| 24 | 21-22 7(f) | Diagram | Si on Ga and As sites in GaAs |
| 25 | 21-22 7(f) | Diagram | GaAs band diagram with $E_F$ below $E_{Fi}$ |
| 26 | 20-21 1(a) | Graph | Fermi-Dirac function at three temperatures |
| 27 | 20-21 1(a) | Diagram | band diagrams of intrinsic, n-type and p-type Si |
| 28 | 20-21 1(a) | Graph | Fermi level against doping at 300 K |
| 29 | 20-21 1(d) | Diagram | band diagram of the As-doped sample |
| 30 | 20-21 1(d) | Graph | ionized fraction and Fermi level against temperature |
| 31 | 20-21 2(a) | Diagram | band diagrams of a compensated semiconductor for three cases |
| 32 | 20-21 2(a) | Graph | Fermi level against acceptor concentration |
| 33 | 20-21 3(a) | Diagram | charge balance in a compensated semiconductor |
| 34 | 20-21 6(b) | Diagram | donor states at 0 K, low temperature and 300 K |
| 35 | 20-21 6(b) | Graph | $n_0/N_d$ against temperature |
| 36 | 20-21 TT1 7(a) | Diagram | donor states at 0 K, low temperature and 300 K |
| 37 | 20-21 TT1 7(b) | Graph | Fermi-Dirac function and donor occupation function |
| 38 | 20-21 TT1 7(d) | Diagram | Si on a Ga site, on an As site and in silicon |
| 39 | 20-21 TT1 7(e) | Graph | $n_0/N_d$ against temperature |
| 40 | 20-21 TT1 7(f) | Graph | density of states, occupation and carrier distributions |
| 41 | 20-21 TT2 8(b) | Diagram | band diagrams of a compensated semiconductor for three cases |
| 42 | 20-21 TT2 8(c) | Diagram | $E_c$, $E_v$, $E_{Fi}$ and $E_F$ of p-type Si, $E_F$ below $E_{Fi}$ near $E_v$ |
| 43 | 20-21 TT2 8(d) | Diagram | Si on a Ga site, on an As site and in silicon |
| 44 | 20-21 TT2 8(e) | Diagram | non-degenerate and degenerate n- and p-type band diagrams |
| 45 | 20-21 TT2 8(f) | Diagram | charge balance in a compensated semiconductor |
| 46 | 19-20 2(a) | Diagram | charge balance in a compensated semiconductor |
| 47 | 19-20 2(c) | Graph | electron concentration against temperature with the three regions |
| 48 | 19-20 2(d) | Diagram | band diagrams of a compensated semiconductor for three cases |
| 49 | 19-20 3(a) | Graph | density of states, Fermi function and carrier distributions of n-type on one energy axis |
| 50 | 19-20 TT1 7(a) | Diagram | charge balance in a compensated semiconductor |
| 51 | 19-20 TT1 7(b) | Graph | Fermi level against doping at 300 K |
| 52 | 19-20 TT1 7(b) | Graph | Fermi level against temperature for several dopings |
| 53 | 19-20 TT1 7(c) | Graph | $n_0$ and $n_i$ at 400 K against $E_g$ |
| 54 | 18-19 1(a) | Graph | electron concentration against temperature with freeze-out, extrinsic and intrinsic regions |
| 55 | 18-19 1(a) | Graph | Fermi level against temperature |
| 56 | 18-19 1(a) | Diagram | donor states at 0 K, low T and 300 K |


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| No. | Paper | Kind | Diagram |
|---|---|---|---|
| 57 | 18-19 7(b) | Graph | density of states, Fermi function and their product near the conduction band |
| 58 | 18-19 TT2 8(b) | Diagram | GaAs band diagram with the Fermi level near the valence band |
| **Carrier transport** | | | |
| 59 | 20-21 2(b) | Diagram | graded doping profile and its equilibrium band diagram with the induced field |
| 60 | 20-21 3(c) | Diagram | electron path with and without a field |
| 61 | 20-21 3(c) | Diagram | lattice and ionized impurity scattering |
| 62 | 20-21 4(a) | Graph | mobility against temperature, log-log |
| 63 | 20-21 5(d) | Diagram | Hall bar with the force balance |
| 64 | 20-21 5(d) | Diagram | carriers piling up in p-type and n-type bars |
| 65 | 19-20 1(c) | Graph | Maxwell-Boltzmann speed distributions for the two masses |
| 66 | 19-20 2(b) | Diagram | Hall bar with dimensions, current, field and Hall field |
| 67 | 19-20 3(d) | Graph | electron mobility against total impurity with the trial points |
| 68 | 19-20 4(a) | Graph | mobility against temperature, log-log |
| 69 | 19-20 4(a) | Graph | electron and hole mobility against doping |
| 70 | 18-19 1(d) | Diagram | Hall bar with current, magnetic field, Hall field and Hall voltage |
| 71 | 18-19 2(c) | Diagram | band diagram of a graded n-type semiconductor with drift and diffusion |
| 72 | 18-19 5(a) | Graph | electron concentration against distance with the flux across $x = 0$ |
| 73 | 18-19 TT2 8(c) | Graph | electron and hole mobility against total impurity concentration |
| **Excess carriers** | | | |
| 74 | 20-21 1(b) | Diagram | diffusion of carriers down a concentration gradient |
| 75 | 20-21 1(b) | Graph | charge density, field and potential of a PN junction |
| 76 | 20-21 3(d) | Graph | excess electrons decaying into the bar |
| 77 | 20-21 3(d) | Graph | total electron and hole concentrations on a log scale |
| 78 | 20-21 5(a) | Diagram | excess electrons and holes held together by their field |
| 79 | 20-21 5(a) | Graph | excess carrier pulse at three times, with and without a field |
| 80 | 20-21 TT1 7(c) | Diagram | n-type sample in equilibrium and under light |
| 81 | 19-20 3(c) | Diagram | net charge drawing in majority carriers |
| 82 | 19-20 3(c) | Graph | exponential decay of the net charge |
| 83 | 19-20 4(b) | Diagram | excess electrons and holes held together by their internal field |
| 84 | 19-20 4(d) | Graph | time decay and build-up of a uniform excess |
| 85 | 19-20 4(d) | Graph | steady-state decay from a surface |
| 86 | 19-20 4(d) | Graph | excess carrier pulse at three times with and without a field |
| 87 | 19-20 5(a) | Graph | excess carriers decaying on both sides of the source |
| 88 | 19-20 5(c) | Diagram | band diagram in equilibrium and with quasi-Fermi levels |
| 89 | 19-20 TT2 8(a) | Graph | excess carriers decaying on both sides of the source |
| 90 | 19-20 TT2 8(b) | Diagram | excess electrons and holes held together by their internal field |
| 91 | 19-20 TT2 8(c) | Diagram | band-to-band, trap (SRH) and Auger processes |
| 92 | 18-19 5(c) | Graph | excess carrier concentration decaying with distance, with $L_p$ marked |
| **PN junction** | | | |
| 93 | 20-21 2(c) | Diagram | PN junction with its space charge region |
| 94 | 20-21 2(c) | Graph | charge density, field and potential across an abrupt junction |
| 95 | 20-21 3(b) | Diagram | p and n band diagrams before contact |
| 96 | 20-21 3(b) | Diagram | equilibrium band diagram of a PN junction with all labels |
| 97 | 20-21 4(d) | Diagram | band diagrams under forward and reverse bias |


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| No. | Paper | Kind | Diagram |
|---|---|---|---|
| 98 | 20-21 4(d) | Graph | hole and electron concentrations on the n side |
| 99 | 20-21 6(d) | Graph | charge density, field and potential across the junction |
| 100 | 19-20 3(b) | Diagram | equilibrium band diagram of a pn junction |
| 101 | 19-20 3(b) | Diagram | drift and diffusion fluxes across the space charge region |
| 102 | 19-20 4(c) | Graph | space charge and field triangles at 300 K and 80 K |
| 103 | 19-20 5(b) | Graph | minority carrier distributions on both sides of a forward-biased junction |
| 104 | 19-20 5(b) | Graph | electron, hole and total current against position |
| 105 | 19-20 5(d) | Graph | field triangles at 0 and 8 V reverse bias |
| 106 | 18-19 2(a) | Graph | diode current and voltage against time when switched from forward to reverse |
| 107 | 18-19 2(a) | Graph | stored minority charge removed during storage time |
| 108 | 18-19 2(d) | Diagram | charge added at the edges of the space charge region for a change in reverse bias |
| 109 | 18-19 2(d) | Graph | junction capacitance against reverse voltage |
| 110 | 18-19 2(d) | Graph | $(1/C)^2$ against reverse voltage |
| 111 | 18-19 4(a) | Graph | electron and hole current densities through a forward-biased pn junction |
| 112 | 18-19 TT2 8(a) | Graph | charge density, electric field and potential across the junction |
| **Tunnel diode** | | | |
| 113 | 20-21 5(c) | Diagram | electron wave at a thick and a thin barrier |
| 114 | 20-21 5(c) | Diagram | tunnel diode band diagrams at several biases |
| 115 | 20-21 5(c) | Graph | tunnel diode I-V curve |
| 116 | 18-19 4(e) | Diagram | tunnel diode band diagrams at zero bias, peak, valley and reverse bias |
| 117 | 18-19 4(e) | Graph | tunnel diode I-V characteristic with peak, valley and negative resistance |
| **Metal-semiconductor contacts** | | | |
| 118 | 19-20 1(d) | Diagram | metal and n-type band diagrams before and after contact |
| 119 | 19-20 1(d) | Diagram | ohmic contact band diagram |
| 120 | 19-20 1(d) | Graph | I-V of rectifying and ohmic contacts |
| 121 | 18-19 4(c) | Diagram | band diagrams of rectifying and ohmic contacts |
| 122 | 18-19 4(c) | Graph | I-V curves of rectifying and ohmic contacts |
| 123 | 18-19 6(d) | Diagram | metal and n-type band diagrams before and after contact |
| 124 | 18-19 6(d) | Diagram | Schottky contact under forward and reverse bias |
| **MOS structure** | | | |
| 125 | 20-21 4(b) | Diagram | MOS capacitor and n-channel MOSFET cross sections |
| 126 | 20-21 4(b) | Diagram | charges in accumulation, depletion and inversion |
| 127 | 20-21 4(b) | Diagram | band diagrams in the three modes |
| 128 | 20-21 6(a) | Diagram | single-gate, planar double-gate and FinFET schematics |
| 129 | 20-21 6(a) | Graph | subthreshold current of single and double gate |
| 130 | 20-21 6(c) | Diagram | band diagrams of a p-type MOS structure from depletion to inversion |
| 131 | 20-21 6(c) | Diagram | charge distribution at threshold inversion |
| 132 | 19-20 6(a) | Diagram | MOS charges at the maximum inversion charge with the Gauss balance |
| 133 | 18-19 3(a) | Diagram | MOS band diagrams at zero, negative, small positive and large positive gate voltage |
| 134 | 18-19 3(b) | Graph | threshold voltage against substrate doping for several oxide thicknesses |
| 135 | 18-19 3(c) | Graph | threshold voltage against source-to-body voltage |
| 136 | 18-19 3(c) | Diagram | depletion region with and without body bias |
| 137 | 18-19 5(b) | Graph | ideal low-frequency C-V curve of a p-type MOS capacitor with the three regions |
| 138 | 18-19 5(b) | Diagram | charge in accumulation, depletion and inversion |
| **Bipolar transistor** | | | |
| 139 | 20-21 4(c) | Diagram | NPN structure and the four modes on the $V_{BE}$–$V_{CB}$ plane |
| 140 | 20-21 4(c) | Graph | minority carrier distributions in the four modes |
| 141 | 20-21 5(b) | Diagram | NPN in forward-active mode with depletion layers |


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| No. | Paper | Kind | Diagram |
|---|---|---|---|
| 142 | 20-21 5(b) | Graph | minority carrier concentrations in emitter, base and collector |
| 143 | 19-20 6(b) | Diagram | four modes on the $V_{BE}$–$V_{BC}$ plane with the cases |
| 144 | 19-20 6(d) | Diagram | minority carriers in emitter, base and collector |
| 145 | 19-20 6(d) | Graph | $\gamma$ against $N_B/N_E$ |
| 146 | 18-19 3(d) | Circuit | common-emitter amplifier |
| 147 | 18-19 3(d) | Graph | input and output waveforms of the amplifier |
| 148 | 18-19 4(b) | Diagram | current components in an npn transistor in active mode |
| 149 | 18-19 4(d) | Graph | minority carrier distribution in the base for two collector voltages |
| 150 | 18-19 4(d) | Graph | collector current against collector-emitter voltage with the Early voltage |
| 151 | 18-19 6(b) | Graph | emitter injection efficiency against the doping ratio |
| 152 | 18-19 6(c) | Diagram | npn cross section with lateral base current and injection crowded at the emitter edges |
| 153 | 18-19 6(c) | Diagram | interdigitated emitter and base fingers |
| **JFET** | | | |
| 154 | 18-19 5(d) | Diagram | n-channel JFET cross section with channel thickness and depletion regions |

# 9. Comparisons:

Every comparison asked, in book order, with the key difference in one line. A pair asked in several papers is one row.

| Pair | Key difference | Asked in |
|---|---|---|
| **Crystal structure** | | |
| Amorphous vs polycrystalline vs single crystal | short-range order only vs ordered grains with boundaries vs one lattice throughout | 19-20 1(a) |
| **Quantum basics** | | |
| Density of states vs carrier concentration | states available vs states filled | 18-19 7(a) |
| Band A vs band B | larger curvature, lighter mass vs flatter, heavier mass | 18-19 7(c) |
| Fermi-Dirac vs Boltzmann | exact, at most one electron per state vs approximation valid for $E - E_F > 3kT$ | 21-22 7(a) |
| **Semiconductor in equilibrium** | | |
| Freeze-out vs intrinsic region | dopants not ionized, $n_0 < N_d$ vs thermal generation, $n_0 \to n_i$ | 18-19 1(a) |
| Complete ionization vs freeze-out | every donor gives its electron, $n_0 = N_d$ vs electrons fall back to donors at low $T$, $n_0 < N_d$ | 20-21 6(b) |
| Degenerate vs non-degenerate | $E_F$ inside a band, metal-like vs $E_F$ in the gap, $n_0 p_0 = n_i^2$ holds | 20-21 TT2 8(e) |
| Donor vs acceptor | gives an electron, level near $E_c$, becomes + vs takes an electron, level near $E_v$, becomes − | 21-22 7(b) |
| Si on a Ga site vs Si on an As site | donor (+) vs acceptor (−) | 21-22 7(f) |
| **Carrier transport** | | |
| Net vs total doping | $N_d - N_a$ sets $n_0$ vs $N_d + N_a$ sets the mobility | 18-19 TT2 8(c) |
| Lattice vs ionized impurity scattering | grows with temperature, $\mu_L \propto T^{-3/2}$ vs weakens with temperature, $\mu_I \propto T^{3/2}/N_I$ | 20-21 3(c), 19-20 4(a) |
| Drift vs diffusion current | driven by a field, $en\mu E$ vs driven by a gradient, $eD\,dn/dx$; linked by $D/\mu = kT/e$ | 20-21 2(b) |
| **Excess carriers** | | |
| Band-to-band vs SRH recombination | direct, radiative, GaAs vs through traps, heat, Si | 19-20 TT2 8(c) |


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| Pair | Key difference | Asked in |
|---|---|---|
| Minority vs majority carriers | fewer, set by $n_i^2/N$, move by diffusion vs more, set by the doping, carry drift current | 20-21 1(b) |
| Equilibrium vs total concentration | $n_0$ fixed by doping and $T$ vs $n = n_0 + \delta n$ out of equilibrium | 20-21 TT1 7(c) |
| **PN junction** | | |
| Storage time vs fall time | stored charge removed at constant $I_R$ vs current decays to 10 % | 18-19 2(a) |
| Minority vs majority current | diffusion, decays over $L$ vs makes up the constant total | 18-19 4(a) |
| **Metal-semiconductor contacts** | | |
| Ohmic vs rectifying contact | linear, either direction vs barrier, one direction | 18-19 4(c) |
| Schottky vs pn diode | majority carriers, no stored charge, lower turn-on vs minority injection, storage time | 18-19 6(d) |
| Schottky barrier vs ohmic contact | depletion, barrier, thermionic emission, rectifying vs accumulation or tunneling, no barrier, linear | 19-20 1(d) |
| **MOS structure** | | |
| Accumulation vs depletion vs inversion | majority carriers gather vs carriers pushed away, ions uncovered vs minority carriers form a channel | 20-21 4(b) |
| Single-gate vs double-gate MOSFET | gate controls one surface, short-channel effects vs gate on both sides, better control, near-ideal swing | 20-21 6(a) |

# 10. Drills:

Every drawing drill asked, in book order: what to draw and the data given. The data changes from paper to paper; the method does not.

| Draw | Data given | Asked in |
|---|---|---|
| **Semiconductor in equilibrium** | | |
| p-type band diagram to scale | $N_a = 10^{16}$ cm$^{-3}$, $E_{Fi} - E_F = 0.35$ eV, $E_F - E_v = 0.18$ eV | 20-21 TT2 8(c) |

# 11. Repeated questions:

14 questions came back, exactly or with new numbers; 2 of them in three or more papers. Most repeated first; the latest paper first in each row.

| No. | Times | Topic | Subtopic | Asked in (marks) |
|---|---|---|---|---|
| 1 | 6 | Semiconductor in equilibrium | Carrier concentrations | 21-22 7(c) (5), 20-21 3(a) (5), 20-21 TT1 7(f) (5), 20-21 TT2 8(f) (5), 19-20 2(a) (6), 19-20 TT1 7(a) (5) |
| 2 | 6 | Quantum basics | Fermi probability | 21-22 7(d) (5), 21-22 7(e), 20-21 2(d) (10), 20-21 TT1 7(h) (5), 20-21 TT2 8(g) (5), 20-21 TT2 8(h) (5) |
| 3 | 2 | Carrier transport | Einstein relation | 20-21 2(b) (5), 18-19 2(c) (6) |
| 4 | 2 | MOS structure | Band diagrams | 20-21 4(b) (5), 18-19 3(a) (8) |
| 5 | 2 | Tunnel diode | Operation | 20-21 5(c) (5), 18-19 4(e) (6) |
| 6 | 2 | PN junction | Depletion width | 20-21 6(d) (10), 18-19 TT2 8(a) (3) |
| 7 | 2 | Semiconductor in equilibrium | Compensation | 20-21 2(a) (5), 19-20 2(d) (6) |
| 8 | 2 | Carrier transport | Mobility | 20-21 4(a) (5), 19-20 4(a) (6) |
| 9 | 2 | Semiconductor in equilibrium | Dopants | 20-21 TT1 7(d) (1), 20-21 TT2 8(d) (1) |
| 10 | 2 | Semiconductor in equilibrium | Temperature dependence | 19-20 2(c) (5), 18-19 1(a) (7) |

| No. | Times | Topic | Subtopic | Asked in (marks) |
|---|---|---|---|---|
| 11 | 2 | PN junction | Ideal diode current | 19-20 5(b) (8), 18-19 4(a) (7) |
| 12 | 2 | Metal-semiconductor contacts | Schottky and ohmic contacts | 19-20 1(d) (6), 18-19 4(c) (4) |
| 13 | 2 | Excess carriers | Ambipolar transport | 19-20 4(b) (7), 19-20 TT2 8(b) (7) |
| 14 | 2 | Excess carriers | Steady state | 19-20 5(a) (7), 19-20 TT2 8(a) (6) |

A sub-part with no marks after it had none printed on its paper.

## 12. Question style:

How each topic was asked, as a number of questions.

| Topic | Theory | Derivation | Numerical | Diagram |
|---|---|---|---|---|
| Crystal structure | 1 | · | 1 | · |
| Quantum basics | 7 | · | 6 | · |
| Semiconductor in equilibrium | 16 | 7 | 8 | 1 |
| Carrier transport | 3 | 4 | 5 | · |
| Excess carriers | 5 | 6 | 2 | · |
| PN junction | 4 | 2 | 5 | 1 |
| Tunnel diode | 2 | · | · | · |
| Metal-semiconductor contacts | 3 | · | · | · |
| MOS structure | 6 | · | 1 | 2 |
| Bipolar transistor | 7 | · | 2 | · |
| JFET | · | · | 2 | · |

## 13. Study order:

Topics by marks offered, with the share of all marks covered so far. Papers offer more marks than you answer, so this is a guide to where marks have been, not a score.

| No. | Topic | Marks | Covered so far |
|---|---|---|---|
| 1 | Semiconductor in equilibrium | 130 | 23 % |
| 2 | Excess carriers | 78 | 37 % |
| 3 | Carrier transport | 75 | 50 % |
| 4 | PN junction | 72 | 63 % |
| 5 | Quantum basics | 58 | 73 % |
| 6 | MOS structure | 54 | 83 % |
| 7 | Bipolar transistor | 47 | 91 % |
| 8 | Metal-semiconductor contacts | 16 | 94 % |
| 9 | JFET | 13 | 96 % |
| 10 | Crystal structure | 12 | 98 % |
| 11 | Tunnel diode | 11 | 100 % |


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